Response to the Reviewers

Manuscript: “Trust-Aware Multi-Agent Reinforcement Learning-Based Cooperative Interception Strategy for Maneuvering Swarms with Target Assignment”

Dear Editor and Reviewers,

We sincerely thank the Editor and the Reviewers for their careful evaluation of our manuscript and constructive comments. We agree that the original manuscript required substantial improvement in presentation, theoretical rigor, and practical validation. In the revised manuscript, we have added a concise notation table, complete pseudocode for IDBO and ART-MAPPO, and intuitive explanations alongside the principal equations, clarified several ambiguous definitions, and systematically reorganized the figures to reduce visual density, with corresponding explanations provided for each figure. The Related Work section has also been updated with 5 relevant journal articles published within the past two years.

We corrected the interpretation of the trust update so that its physical description is consistent with the smoothing process, and rectified the reported episode horizon and engagement-region parameters. The evaluation environment has been reformulated as a three-dimensional hardware-in-the-loop platform incorporating communication delay, packet loss, and sensing uncertainty. All experiments reported in the revised manuscript were conducted on this HIL platform. We further added controlled component ablations, sensitivity analyses of the IDBO assignment weights, reward weights, and training hyperparameters, as well as robustness studies covering partial interceptor failures, unsuccessful and delayed engagements, and an unseen bang-bang maneuver pattern. The IDBO convergence conditions, computational complexity, and communication-delay-dependent scalability have been expanded. For ART-MAPPO, we added a comparison with conventional MAPPO regarding convergence guarantees, exploration–exploitation balance, and policy-update robustness. We believe these revisions substantially improve the clarity, technical soundness, experimental completeness, and practical credibility of the manuscript. All corresponding changes are highlighted in the revised version.

Reviewer 1

Comment 1.1

Reviewer comment 1.1

Provide a concise notation table summarizing key variables and symbols used throughout the paper.

Response. We sincerely thank the reviewer for this helpful suggestion. We agree that the principal symbols were previously defined across several equations, making it inconvenient for readers to locate their meanings. We have therefore added a dedicated appendix entitled “Summary of Key Variables and Symbols.”

The symbols are organized into three groups corresponding to the engagement model, IDBO target assignment, and ART-MAPPO cooperative guidance. The complete addition is reproduced below.

Location of the revision. Appendix A (*Summary of Key Variables and Symbols*), immediately following the reference list.

Appendix A: Summary of Key Variables and Symbols

Table IX consolidates the principal variables and symbols used throughout the paper.

Table IX
Key Variables and Symbols

Symbol	Description	Symbol	Description
Engagement model			
(x, y, z)	UAV position coordinates	V, γ, θ	Speed, heading, and flight-path angle
$\mathbf{p}_A, \mathbf{p}_D$ $\mathbf{v}_A, \mathbf{v}_D$	Attacker/defender positions and velocities	$\mathbf{r}_{AD}, \boldsymbol{\ell}_{AD}$	Relative-position vector and LOS unit vector
$r, \dot{r}$	Relative range and range rate	q^γ, q^θ	LOS azimuth and elevation
n_x, n_y, n_z	Axial, lateral, and normal accelerations	t_{go}	Estimated time-to-go
ZEM	Zero-effort-miss distance		
IDBO target assignment			
$\mathcal{U}, \mathcal{T}$	Interceptor and target sets	M, N	Numbers of interceptors and targets
X_{ij}	Binary assignment of u_i to t_j	P_{ij}	Composite interception probability
$w_{1,\text{IDBO}}$ $w_{2,\text{IDBO}}$	Reachability and geometry weights	$L_{\max}$	Per-target interceptor capacity
$\mathbf{x}_i$	Target-preference vector of u_i	$\mathcal{F}_i$	Local IDBO fitness
A_{ij}	Adversarial advantage for pair (i, j)	Ψ_{ij}	Assignment-saturation penalty
$\mathcal{N}_i$	Communication-neighbor set of u_i	c_{ij} $\mathcal{Z}_j$	Bid and winner list for target t_j
c_t	Common IDBO decay schedule	$\Gamma^{(k)}$ ϵ_{IDBO}	Swarm disagreement and convergence tolerance
ART-MAPPO cooperative guidance			
$\mathbf{o}_i^{(t)}$	Local observation of interceptor i	s_t	Centralized global state
$\mathbf{A}_j$	Attention matrix of head j	$\mathbf{c}_t$	GRU hidden state
$\pi_{\boldsymbol{\vartheta}_i}$	Decentralized actor policy	$V_\phi(s_t)$	Centralized critic value
$\mathbf{a}_i$	Three-axis acceleration command	$\mathcal{T}_i^{(k)}$	Smoothed trust level
$\tilde{R}_i^{(k)}$	Normalized relative return	$\beta(\mathcal{T}_i)$ π^i_{guided}	Guided-policy weight and distribution
$\hat{A}_t$	GAE advantage estimate	$r_t(\boldsymbol{\vartheta})$ ϵ	Probability ratio and PPO clip threshold
J^{DC}, c	Dual-Clip objective and floor	$\hat{D}_{\text{KL}}$ $\beta_{\text{KL}}, \delta_{\text{targ}}$	KL estimate, penalty coefficient, and target
$r_i^{(t)}$	Per-step composite reward	$w_1, \ldots, w_5$	Reward-component weights
$\mathcal{G}_i$	Same-target interceptor group	$\bar{t}_{go}^{(t)}$	Group mean time-to-go
$L_V(\phi)$	Centralized critic value loss	$\mathcal{L}(\boldsymbol{\vartheta}, \phi)$	Overall training loss

Comment 1.2

Reviewer comment 1.2

Include pseudocode or algorithm boxes for IDBO and ART-MAPPO to improve reproducibility.

Response. We sincerely thank the reviewer for this valuable suggestion. We agree that the equations in the original manuscript defined the individual components of IDBO and ART-MAPPO, but did not provide sufficiently compact procedural descriptions for reproducing the complete algorithms. We have therefore added two pseudocode boxes to the revised manuscript.The two added algorithms are reproduced below for the reviewer's convenience.

(1) IDBO target-assignment pseudocode.

Location of the revision. Section III-B (*Distributed Consensus-Based IDBO Framework*), immediately after the convergence criterion in Eq. (26) and before the per-UAV complexity analysis in Eq. (27).

Algorithm 1 IDBO Target Assignment

Require: Interceptor and target sets $\mathcal{U}, \mathcal{T}$; engagement states; communication graph; capacity $L_{\max}$; local population size N_{pop}; local search horizon T; tolerance ϵ_{IDBO}

Ensure: Feasible consensus assignment $\mathbf{X}^*$

Each interceptor u_i initializes a local preference population, assignment replica, bid/winner records, and timestamps; set $k \leftarrow 0$

repeat

    Evaluate $\mathbf{P}$ using Eq. (8) and the adversarial advantages using Eq. (15)

    for $t = 1$ to T **do**

        Set the linearly decaying coefficients $c_t = c_0(1 - t/T)$ for the four IDBO operators

        Evaluate all local candidates with $\mathcal{F}_i$ in Eq. (13)

        Generate candidates using the rolling, dancing, breeding, and stealing operators according to Eqs. (16)–(19)

        Project the candidate preferences onto $[0,1]^N$ and retain the current best candidate unless a strictly better candidate is obtained

    end for

    Select and binarize the best local preference using Eq. (20); compute the target bids using Eq. (21)

    Exchange local records with $\mathcal{N}_i$, update the top-$L_{\max}$ winner lists using Eq. (22), and resolve assignment conflicts subject to the constraints in Eq. (11)

    Update the local feasible assignment replica and adversarial information; evaluate consensus disagreement; $k \leftarrow k + 1$

until the disagreement falls below ϵ_{IDBO}

Set $\mathbf{X}^*$ to the common feasible assignment replica

return $\mathbf{X}^*$

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(2) ART-MAPPO training pseudocode.

Location of the revision. Section IV-C (*Robust Optimization via Dual-Clip and Adaptive KL*), immediately after the ART-MAPPO workflow in Fig. 5 and before Section IV-D (*Mechanistic Comparison with Conventional MAPPO*).

Algorithm 2 ART-MAPPO for Cooperative Swarm Interception

Require: Actor parameters $\{\boldsymbol{\vartheta}_i\}_{i=1}^{M}$; centralized critic parameters ϕ; episode horizon T; training episodes K; PPO clip parameter ϵ; dual-clip floor c; target KL divergence δ_{targ}; trust parameters α_T, ρ, and τ_T; optimizer parameters

Ensure: Trained decentralized policies $\{\pi_{\boldsymbol{\vartheta}_i}\}_{i=1}^{M}$

Initialize $\{\boldsymbol{\vartheta}_i\}_{i=1}^{M}$ and ϕ; set $\mathcal{T}_i \leftarrow 0.5$ for each interceptor; initialize β_{KL} and the running return statistics μ_R and σ_R^2
for episode $k = 1$ to K **do**
  Reset the environment, clear the rollout buffer, and reset $\mathbf{c}_{i,0}$ for each interceptor
  Set $\boldsymbol{\vartheta}_{i,\mathrm{old}} \leftarrow \boldsymbol{\vartheta}_i$ for all $i = 1, \ldots, M$
  // Decentralized rollout collection
  for $t = 0$ to $T - 1$ **do**
    for all interceptors $i = 1, \ldots, M$ in parallel **do**
      Encode $\mathbf{o}_i^{(t)}$ using the attention–residual network according to Eqs. (28)–(34), and update the temporal state $\mathbf{c}_{i,t}$ using the GRU in Eq. (36)
      Form the policy $\pi_{\boldsymbol{\vartheta}_i}(\cdot \mid \mathbf{o}_i^{(t)})$ according to Eq. (37), and sample $\mathbf{a}_{i,t}$ according to Eqs. (43)–(44), with $\beta(\mathcal{T}_i) = 1 - \mathcal{T}_i$
      Clip $\mathbf{a}_{i,t}$ to the three-dimensional feasible action envelope $\mathcal{A}$
    end for
    Step the environment once using the joint action $\mathbf{a}_t = (\mathbf{a}_{1,t}, \ldots, \mathbf{a}_{M,t})$ and store the joint transition in the rollout buffer
  end for
  // Trust update based on episode returns
  Update μ_R and σ_R^2 using Eqs. (39)–(40), and compute the normalized return $\widetilde{R}_i^{(k)}$ using Eq. (41)
  Update the trust level $\mathcal{T}_i^{(k+1)}$ using Eq. (42)
  // Centralized optimization using the global state defined in Eq. (38)
  Compute the GAE advantages $\hat{A}_t$ and value targets using Eqs. (46)–(47)
  for each mini-batch update **do**
    Evaluate the dual-clip surrogate objective $J^{\mathrm{DC}}(\boldsymbol{\vartheta})$ using Eqs. (48)–(49)
    Estimate $\widehat{D}_{\mathrm{KL}}$ using Eq. (51), and compute the value loss $L_V(\phi)$ using Eq. (53)
    Update $\{\boldsymbol{\vartheta}_i\}_{i=1}^{M}$ and ϕ by minimizing the total loss $\mathcal{L}(\boldsymbol{\vartheta}, \phi)$ in Eq. (54) using AdamW with cosine learning-rate scheduling
  end for
  Adapt the KL weight $\beta_{\mathrm{KL}}^{(k+1)}$ using Eq. (52)
end for
return $\{\pi_{\boldsymbol{\vartheta}_i}\}_{i=1}^{M}$

Comment 1.3

Reviewer comment 1.3

Add intuitive explanations before or alongside major equations, especially in Sections III and IV.

Response. We sincerely thank the reviewer for this careful and constructive suggestion. We have therefore added concise explanations immediately alongside eight groups of major equations. The corresponding additions are reproduced in blue in the white boxes.

(1) Location of the revision. Section III-B (*Distributed Consensus-Based IDBO Framework*), immediately after Eqs. (13)–(14). We added an intuitive interpretation of how the fitness function combines target preference with the saturation penalty.

In intuitive terms, each interceptor increases its preference for targets that combine high interception likelihood with a strong tactical advantage, while the saturation penalty redirects the local search when neighboring preferences indicate that a target has reached its allocation capacity.

(2) Location of the revision. Section III-B, in Eq. (15) and the paragraph immediately following it. The geometric factor is expressed through the normalized inner product between the interceptor velocity and the target LOS, and its physical meaning in the three-dimensional engagement is stated explicitly.

$$
\begin{aligned}
A_{ij} = {} & \exp\left(-\frac{\Delta t_{ij}}{T_{\max}}\right)\left(1 - \frac{\|\mathbf{v}_i - \mathbf{v}_j\|_2}{v_{\max}}\right) \\
& \cdot \max\left(0, \frac{\mathbf{v}_i^\top(\mathbf{p}_j - \mathbf{p}_i)}{\|\mathbf{v}_i\|_2\,\|\mathbf{p}_j - \mathbf{p}_i\|_2}\right) \\
& - \sum_{k \neq i} \frac{P_{kj}}{P_{ij} + P_{kj}} \cdot \sigma(\hat{x}_{kj}).
\end{aligned}
$$

$\mathbf{p}_i = [x_i, y_i, z_i]^\top$ and $\mathbf{v}_i = [\dot{x}_i, \dot{y}_i, \dot{z}_i]^\top$ denote the three-dimensional position and velocity vectors of interceptor u_i, while $\mathbf{p}_j = [x_j, y_j, z_j]^\top$ and $\mathbf{v}_j = [\dot{x}_j, \dot{y}_j, \dot{z}_j]^\top$ denote those of target t_j.

The normalized inner product is the cosine of the three-dimensional angle between the interceptor velocity and the LOS to target t_j; the maximum operator assigns zero geometric advantage when the target lies outside the interceptor's forward hemisphere. The three multiplicative factors encode temporal, energetic, and geometric advantage, respectively, while the subtracted term penalizes competition from other interceptors with high preferences for the same target.

(3) Location of the revision. Section III-B, immediately after the rolling, dancing, breeding, and stealing operators in Eqs. (16)–(19), and before the thresholding equation. We explained how the decaying coefficients progressively suppress the stochastic components of the four update operators.

The coefficients $\alpha, \beta, \gamma_{\text{IDBO}}, \delta, \delta', \eta_{\text{IDBO}}$, together with the dancing-noise scale σ_d, follow the common linear schedule $c_t = c_0(1 - t/T)$, where T denotes the local IDBO search horizon. During each local search, the candidate preferences are projected onto $[0, 1]^N$, and the best-so-far candidate is retained unless a strictly better candidate is obtained. The coefficient schedule progressively attenuates both the directed update terms and the stochastic perturbations. In particular, the rolling perturbation $\beta\boldsymbol{\xi}_i \mathrm{e}^{-t/T}$ and the dancing perturbation $\mathbf{n}_i \sim \mathcal{N}(0, \sigma_d^2\mathbf{I})$ vanish as t approaches T, suppressing late-stage fluctuations in the preference population. The schedule therefore satisfies the vanishing-gain condition $c_t \to 0$ used in Robbins–Monro-type stochastic approximation [51]. At $c_T = 0$, the four update operators reduce to the identity map, while elitist retention prevents the incumbent from being replaced by an inferior candidate. Consequently, once no strictly improving candidate is generated, the retained solution remains invariant and constitutes a stable fixed point of the implemented local search.

(4) Location of the revision. Section III-B, immediately after the neighborhood-consensus update in Eq. (22). We added a concise explanation connecting the thresholding, bidding, and winner-list updates in Eqs. (20)–(22).

Equations (20)–(22) convert the continuous preferences into local assignment candidates. Each consensus update exchanges the current bids among neighboring interceptors and retains at most the $L_{\max}$ strongest bids for each target. Once the local search has stabilized and the bids no longer change, repeated consensus updates propagate the winner information through the connected network and bring the local winner lists into agreement.

(5) Location of the revision. Section IV-A (*Enhanced Situational Awareness via Attention-Residual Network*), immediately after the multi-head attention output in Eq. (31). We clarified the interpretation of the attention matrix in Eq. (30) and the role of the output projection in Eq. (31).

Each row of $\mathbf{A}_j$ specifies how the observable kinematic features contribute to the update of one feature token. After the head outputs are concatenated and flattened, $\mathbf{W}^O$ maps the attended information from all feature tokens back to the d-dimensional latent space for residual fusion with $\mathbf{h}^{(0)}$.

(6) Location of the revision. Section IV-A, immediately after the residual-block updates in Eqs. (32)–(34), and before the Jacobian analysis in Eq. (35). We explained the distinct functions of the nonlinear branch and the identity shortcut.

The nonlinear branch refines the attended representation, while the shortcut carries the incoming engagement features directly to the next block. This structure reduces the need to reconstruct unchanged kinematic information at every layer.

(7) Location of the revision. Section IV-B (*Trust-Modulated Tactical Exploration*), immediately after the trust update in Eq. (42). We added an intuitive explanation of the bounded performance reference, the smoothing operation, and the resulting adjustment of exploration.

The sigmoid term maps the normalized relative return $\tilde{R}_i^{(k)}$ to a bounded performance reference in $(0, 1)$. Equation (42) combines the previous trust level and the current performance reference with the weights $1 - \alpha_T$ and α_T, respectively, yielding a smoothed trust estimate. The smoothing process tracks changes in relative performance while attenuating episode-to-episode return fluctuations. Sustained above-average returns keep the performance reference above 0.5 and bias the smoothed trust estimate toward a correspondingly higher level, thereby reducing the probability assigned to the guided exploratory policy; sustained below-average returns produce the opposite allocation.

(8) Location of the revision. Section IV-C (*Robust Optimization via Dual-Clip and Adaptive*

KL), immediately after the overall loss in Eq. (54). We clarified which loss terms update the actor and critic and how the remaining terms regulate policy change and exploration.

Minimizing this loss updates the actor through the Dual-Clip objective, while the KL and entropy terms regulate policy change and exploration. The value-loss term trains the centralized critic.

Comment 1.4

Reviewer comment 1.4

Expand discussion on real-world applicability, including communication delays, sensing uncertainty, and 3D dynamics.

Response. We sincerely thank the reviewer for this important and constructive suggestion. We agree that the original planar formulation and idealized simulation description did not adequately reflect the operating conditions of practical cooperative interception. In response, we reformulated the engagement model in three-dimensional space, incorporated delayed and noisy state information into the observation process, and implemented the complete target-assignment and cooperative-guidance framework on a hardware-in-the-loop (HIL) platform. All experiments reported in the revised manuscript were rerun on this updated three-dimensional HIL platform under the stated communication and sensing conditions; the corresponding results are presented in Section V (*Simulation Verification*). The specific revisions are detailed below.

(1) Three-dimensional engagement dynamics.

Location of the revision. Section II-A (*Problem Modeling*), Eqs. (1)–(3) and (5). The planar kinematic, relative-motion, acceleration, and ZEM formulations were replaced by their three-dimensional counterparts.

The object of study in this paper is the fixed-wing UAV, and the interception problem is reduced to a two-dimensional horizontal plane formulated in three-dimensional space.

$$\begin{cases} \dot{x} = V\cos\theta\cos\gamma, \\ \dot{y} = V\cos\theta\sin\gamma, \\ \dot{z} = V\sin\theta, \end{cases} \tag{1}$$

(x, y, z) are the three-dimensional position coordinates, V is the velocity magnitude, γ is the heading angle, and θ is the flight-path angle.
Let $\mathbf{p}_A, \mathbf{p}_D \in \mathbb{R}^3$ and $\mathbf{v}_A, \mathbf{v}_D \in \mathbb{R}^3$ denote the attacker and defender states. Their three-dimensional relative geometry is

$$\begin{aligned} &\mathbf{r}_{AD} = \mathbf{p}_A - \mathbf{p}_D, \qquad r = \|\mathbf{r}_{AD}\|_2, \\ &\boldsymbol{\ell}_{AD} = \mathbf{r}_{AD}/r, \qquad \dot{r} = \boldsymbol{\ell}_{AD}^{\top}(\mathbf{v}_A - \mathbf{v}_D). \end{aligned} \tag{2}$$

where r and $\boldsymbol{\ell}_{AD}$ denote the three-dimensional relative distance and LOS unit vector, respectively. As illustrated in Fig.2, q_{ij}^{γ} and q_{ij}^{θ} are the corresponding LOS azimuth and elevation angles.

$$\begin{cases} \dot{V} = n_x g, \\ \dot{\gamma} = \dfrac{n_y g}{V \cos\theta}, \\ \dot{\theta} = \dfrac{n_z g}{V}, \end{cases} \tag{3}$$

where n_x, n_y, and n_z denote the axial, lateral, and normal accelerations, and g denotes gravitational acceleration.

$$\text{ZEM} = \|\mathbf{r}_{AD} + (\mathbf{v}_A - \mathbf{v}_D)t_{go}\|_2 \,. \tag{5}$$

(2) Communication delay and sensing uncertainty.

Location of the revision. Section V-A (*Experimental Setup and Implementation Details*), in the newly added paragraph entitled *Communication Delay and Sensing Uncertainty*, including Eq. (71). Delayed and noisy Cartesian states are now used to construct the defender observations, and communication packet loss is included in the information exchange process.

1) Communication Delay and Sensing Uncertainty: Communication and sensing nonidealities are incorporated into the three-dimensional simulation. The position and velocity measurements used to construct the defender observations are delayed and corrupted by independent zero-mean Gaussian noise,

$$\widehat{\mathbf{p}}(t) = \mathbf{p}(t - \tau_s) + \boldsymbol{\eta}_p, \qquad \widehat{\mathbf{v}}(t) = \mathbf{v}(t - \tau_s) + \boldsymbol{\eta}_v, \tag{71}$$

where $\boldsymbol{\eta}_p \sim \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I}_3)$ and $\boldsymbol{\eta}_v \sim \mathcal{N}(\mathbf{0}, \sigma_v^2 \mathbf{I}_3)$. The nominal settings are $\tau_s = 50$ ms, $\sigma_p = 1$ m, and $\sigma_v = 0.3$ m/s. All derived observation variables, including the LOS geometry, closing speed, and time-to-go, are recomputed from the delayed noisy Cartesian states. The communication process is subject to a 50 ms communication delay and a 1% packet-loss rate, with the latest valid state packet retained when a packet is lost.

(3) HIL Platform.

Location of the revision. Section V-A, in the newly added paragraph entitled *HIL Platform*, immediately following Eq. (71), together with the platform architecture shown in Fig. 7. The target-assignment and decentralized policy-inference modules are executed on independent embedded computers and communicate with the engagement simulator through a physical network.

2) HIL Platform: As shown in Fig.7, the three-dimensional interception experiment is further implemented on a HIL platform. The policies of defenders D_0–D_4 are deployed on five independent NVIDIA Jetson NX computers, with one defender assigned to each node. The corresponding target-assignment algorithm is also executed on the Jetson NX nodes. The engagement dynamics, attacker cluster, and software policies of the remaining defenders D_5–D_n are executed on the simulation server. The server and Jetson NX nodes exchange observations, actions, and target-assignment results through a physical TCP Ethernet network under the communication conditions described above.

(4) Application of the revised environment in the end-to-end experiments.

Location of the revision. Section V-D (*Cooperative Interception against Maneuvering Targets*), at the beginning of the subsection. We clarified that the complete target-assignment and cooperative-guidance process is evaluated under the three-dimensional dynamics and communication and sensing conditions described above.

We evaluate the complete target-assignment and terminal cooperative-guidance process against two maneuvering scenarios, each benchmarked against classical PN. The experiments are conducted on the HIL platform with communication delays, sensing uncertainty and three-dimensional dynamics. The target-assignment and cooperative-guidance modules are executed sequentially to provide end-to-end validation of the complete interception process.

Comment 1.5

Reviewer comment 1.5

Discuss the sensitivity of the reward function with respect to weights and hyperparameters.

Response. We sincerely thank the reviewer for this important suggestion. We have therefore added two complementary sensitivity studies in Section V-G. The complete additions are reproduced below.

(1) ART-MAPPO reward-weight sensitivity.

Location of the revision. Section V-G (*Parameter Sensitivity Analysis*), in the newly added paragraph *2) ART-MAPPO Reward-Weight Sensitivity* and Table VII.

2) ART-MAPPO Reward-Weight Sensitivity. The sensitivity of the reward design is evaluated in Case 1 using a one-at-a-time perturbation of the five weights in (62). Each weight is scaled by $s \in 0.50, 0.75, 1.00, 1.25, 1.50$, while the remaining weights are fixed at their nominal values. Table VII reports the resulting ISR, temporal-coordination error, terminal lateral acceleration, interception time, and miss distance. The results reveal clear and physically interpretable metric–weight relationships. Increasing w_1 improves terminal accuracy but requires moderately higher terminal lateral acceleration, whereas w_2 has a comparatively mild effect and mainly refines the interception geometry. The terminal-hit weight w_3 primarily governs interception reliability; reducing it to half of its nominal value decreases ISR to 93.8%, while further increasing it provides only marginal reliability improvement at the cost of higher lateral acceleration. The coordination weight w_4 most strongly affects $E_{co\text{-}time}$, which decreases from 0.0830 s to 0.0375 s as the weight increases. In contrast, w_5 directly regulates control effort, reducing E_n from 0.0910 g to 0.0560 g, with a moderate loss in terminal accuracy and interception efficiency. Overall, the nominal configuration provides a balanced operating point among interception reliability, terminal accuracy, temporal coordination, and control economy.

Table VII. Sensitivity Analysis of ART-MAPPO Reward-Function Weighting Coefficients.

Weight	s	ISR (%) ↑	$E_{co\text{-}time}$ (s) ↓	E_n (g) ↓	E_t (s) ↓	E_{miss} (m) ↓
w_1 (dist.)	0.50	96.2	0.0568	0.0675	30.120	3.420
	0.75	97.8	0.0530	0.0696	29.350	3.170
	1.00	**98.7**	**0.0505**	**0.0718**	**28.843**	**2.997**
	1.25	98.9	0.0493	0.0750	28.340	2.860
	1.50	98.8	0.0490	0.0808	28.100	2.830
w_2 (head.)	0.50	97.1	0.0528	0.0785	29.250	3.340
	0.75	98.0	0.0517	0.0748	29.020	3.130
	1.00	**98.7**	**0.0505**	**0.0718**	**28.843**	**2.997**
	1.25	98.8	0.0500	0.0698	28.750	2.890
	1.50	98.7	0.0502	0.0689	28.760	2.870
w_3 (hit)	0.50	93.8	0.0585	0.0645	30.500	3.580
	0.75	97.0	0.0538	0.0687	29.450	3.230
	1.00	**98.7**	**0.0505**	**0.0718**	**28.843**	**2.997**
	1.25	99.0	0.0518	0.0780	28.400	2.900
	1.50	99.1	0.0540	0.0855	28.100	2.870
w_4 (coord.)	0.50	96.8	0.0830	0.0670	28.300	3.150
	0.75	98.0	0.0630	0.0695	28.550	3.060
	1.00	**98.7**	**0.0505**	**0.0718**	**28.843**	**2.997**
	1.25	98.9	0.0430	0.0750	29.050	3.020
	1.50	98.8	0.0375	0.0810	29.550	3.100
w_5 (energy)	0.50	98.9	0.0495	0.0910	28.350	2.880
	0.75	98.8	0.0500	0.0800	28.550	2.930
	1.00	**98.7**	**0.0505**	**0.0718**	**28.843**	**2.997**
	1.25	98.3	0.0518	0.0630	29.200	3.120
	1.50	97.5	0.0545	0.0560	29.850	3.350

(2) ART-MAPPO training-hyperparameter sensitivity.

Location of the revision. Section V-G, in the newly added paragraph *3) Training-Hyperparameter Sensitivity* and Table VIII.

3) Training-Hyperparameter Sensitivity. The initial learning rate η_0, PPO clipping factor ϵ, and target KL divergence δ_{targ} are independently varied to evaluate the sensitivity of ART-MAPPO to its principal training hyperparameters. As shown in Table VIII, the performance remains comparatively stable around the nominal settings, whereas larger deviations generally reduce the Training AUC and final reward. The learning rate shows the most pronounced sensitivity, while the clipping factor maintains similar performance over $\epsilon \in [0.15, 0.25]$. Smaller target-KL values lead to more conservative learning, whereas an overly permissive target KL degrades both metrics. These results indicate that ART-MAPPO is not overly sensitive to moderate hyperparameter variations and that the nominal configuration provides a robust balance between learning efficiency and converged performance.

Table VIII. Sensitivity analysis of ART-MAPPO training hyperparameters.

Hyperparameter	Value	Training AUC (10^4) ↑	Final reward ↑
Initial learning rate η_0	1×10^{-4}	6.656	8.850
	2×10^{-4}	7.111	8.931
	$\mathbf{3 \times 10^{-4}}$	**7.354**	**8.958**
	6×10^{-4}	6.979	8.770
	1×10^{-3}	6.331	8.438
PPO clipping factor ϵ	0.10	7.014	8.871
	0.15	7.354	8.958
	0.20	**7.354**	**8.958**
	0.25	7.310	8.930
	0.30	7.110	8.820
Target KL δ_{targ}	0.010	7.080	8.998
	0.015	7.230	8.978
	0.020	**7.354**	**8.958**
	0.030	7.160	8.868
	0.040	6.940	8.728

Comment 1.6

Reviewer comment 1.6

Provide rigorous discussion on the convergence claims and equilibrium properties.

Response. We sincerely thank the reviewer for this important and constructive comment. In response, we have expanded the discussion for both IDBO and ART-MAPPO. The corresponding revisions are detailed below.

(1) Convergence and fixed-point analysis of IDBO.

Response. We added a rigorous discussion of the convergence properties of IDBO. The additions are reproduced below.

(i) Local-search limiting behavior.

Location of the revision. Section III-B (*Distributed Consensus-Based IDBO Framework*), immediately after the rolling, dancing, breeding, and stealing operators in Eqs. (16)–(19), and before the advantage-adaptive thresholding operation in Eq. (20). We added the following analysis of the coefficient schedule, bounded preference domain, elitist retention, and local-search fixed point:

The coefficients $\alpha, \beta, \gamma_{\text{IDBO}}, \delta, \delta', \eta_{\text{IDBO}}$, together with the dancing-noise scale σ_d, follow the common linear schedule $c_t = c_0(1 - t/T)$, where T denotes the local IDBO search horizon. During each local search, the candidate preferences are projected onto $[0, 1]^N$, and the best-so-far candidate is retained unless a strictly better candidate is obtained. The coefficient schedule progressively attenuates both the directed update terms and the stochastic perturbations. In particular, the rolling perturbation $\beta \boldsymbol{\xi}_i \mathrm{e}^{-t/T}$ and the dancing perturbation $\mathbf{n}_i \sim \mathcal{N}(0, \sigma_d^2 \mathbf{I})$ vanish as t approaches T, suppressing late-stage fluctuations in the preference population. The schedule therefore satisfies the vanishing-gain condition $c_t \to 0$ used in Robbins–Monro-type stochastic approximation [51]. At $c_T = 0$, the four update operators reduce to the identity map, while elitist retention prevents the incumbent from being replaced by an inferior candidate. Consequently, once no strictly improving candidate is generated, the retained solution remains invariant and constitutes a stable fixed point of the implemented local search.

(ii) Relationship among thresholding, bidding, and consensus.

Location of the revision. Section III-B, immediately after the top-$L_{\max}$ winner-list update in Eq. (22), and before the three-phase iteration in Eqs. (23)–(25). We clarified that the consensus statement applies after the local preferences and bids have stabilized:

> Equations (20)–(22) convert the continuous preferences into local assignment candidates. Each consensus update exchanges the current bids among neighboring interceptors and retains at most the $L_{\max}$ strongest bids for each target. Once the local search has stabilized and the bids no longer change, repeated consensus updates propagate the winner information through the connected network and bring the local winner lists into agreement.

(iii) Consensus fixed point and assignment feasibility.

Location of the revision. Section III-B, immediately after the swarm-disagreement convergence criterion in Eq. (26) and before the complexity analysis in Eq. (27). We deleted the previous unqualified claims of monotonic fitness improvement, an $N \cdot D$ iteration bound, and an approximate Nash equilibrium. They have been replaced by the following conditional fixed-point analysis:

> The local optimization and consensus phases have distinct limiting properties. The diminishing coefficient schedule and elitist retention stabilize the local search as described above. For the consensus phase, suppose that the bids remain fixed, the communication graph is connected with diameter D, messages are delivered with bounded delay, and equal bids are resolved deterministically. Under these conditions, the top-$L_{\max}$ winner information propagates by at least one communication hop in each successful round. Consequently, all interceptors obtain the same winner list for each target after at most D successful communication rounds. Bounded delay increases the elapsed consensus time but does not alter the resulting winner lists. Once agreement is reached, further applications of Eq. (22) leave the winner lists unchanged; hence, they form a consensus fixed point. Each list contains at most $L_{\max}$ interceptors, and, provided that a feasible assignment exists, the subsequent conflict-resolution step enforces the constraints in Eq. (11). Here, ϵ_{IDBO} specifies the numerical tolerance for winner-list disagreement.

(2) Convergence analysis and empirical validation of ART-MAPPO.

Response. We added a mechanism-based discussion of the convergence characteristics of ART-MAPPO and a component-ablation study. The additions are reproduced below.

Location of the revision. Section IV-D (*Mechanistic Comparison with Conventional MAPPO*), in the newly added "Convergence analysis" paragraph. The revised text reads:

> *1) Convergence analysis.* Conventional MAPPO primarily relies on centralized value estimation, GAE, and a fixed clipped surrogate to support policy learning. In ART-MAPPO, the attention–residual encoder preserves the coupled kinematic information and maintains a direct gradient path, while the GRU incorporates maneuver history into the policy represen-

tation. This reduces the sensitivity of policy learning to instantaneous state variations. The smoothed trust variable changes the guided-exploration weight gradually according to relative return, allowing the sampling policy to shift steadily toward exploitation as effective cooperative behavior develops. During parameter optimization, Dual-Clip limits anomalous negative surrogate contributions, and adaptive KL regularization adjusts the penalty according to the observed policy deviation. The resulting learning process exhibits lower sampling and update variability, promoting a more stable approach to the steady training regime than conventional MAPPO.

Location of the revision. Section V-E, new subsection *Component Ablation Study.* We added one-at-a-time ablations of the trust-aware mechanism, GRU temporal encoder, and attention–residual backbone. The corresponding additions are reproduced below:

To isolate the independent contribution of each proposed component, we construct three one-at-a-time ablation controls alongside the complete ART-MAPPO model. *"w/o trust-aware mechanism"* disables the trust-modulated fusion of the analytical guidance prior and the learned action; *"w/o GRU temporal encoder"* replaces recurrent state encoding with a memoryless feedforward layer; *"w/o attention–residual backbone"* substitutes a parameter-matched feedforward network while retaining the trust and GRU modules. All variants share the same reward, dynamics, optimiser, training seed, and budget, and are evaluated in 1000 frozen-policy Case 1 Monte Carlo trials. Table V summarises the results. Critically, the three controls exhibit *module-specific* rather than uniformly ordered degradation, providing direct evidence that each component carries a distinct and non-redundant role.

Table V. Component ablation of ART-MAPPO: 1000-trial Case 1 Monte Carlo evaluation.

Variant	Training AUC (10^4) ↑	Final reward ↑	ISR (%) ↑	$E_{co\text{-}time}$ (s) ↓	E_n (g) ↓	E_t (s) ↓	E_{miss} (m) ↓
Full ART-MAPPO	7.354	8.958	**98.7**	0.0505	0.0718	28.843	2.997
w/o trust-aware mechanism	6.037	8.683	97.2	0.0545	0.0739	29.132	3.087
w/o GRU temporal encoder	6.388	8.134	95.8	0.0732	0.0782	30.574	3.297
w/o attention–residual backbone	6.651	8.353	96.1	0.0606	0.0847	29.853	3.507

(i) Trust-aware mechanism. Removing trust produces the largest training-AUC reduction among the three controls (-17.9%, from 7.354×10^4 to 6.037×10^4), yet the final reward falls by only 3.1% and ISR drops by 1.5 pp. The pronounced gap between cumulative-AUC loss and end-of-training reward confirms that the trust-aware mechanism primarily accelerates learning efficiency throughout training, not merely raises the asymptotic policy ceiling—consistent with its design role of dynamically regulating exploration–exploitation balance.

(ii) GRU temporal encoder. The GRU removal yields the lowest final reward (-9.2%) and the largest ISR degradation (-2.9 pp) among all controls. Most distinctively, $E_{co\text{-}time}$ increases by 45% (from 0.0505 s to 0.0732 s), far exceeding the 6% increase in E_t. Without temporal history, the policy can no longer reliably track target manoeuvres or reconcile interceptor time-to-go estimates, leading to simultaneous degradation in both final policy quality and cooperative timing precision.

(iii) Attention–residual backbone. Removing the attentive backbone yields the largest increases in E_n (+18%, from 0.0718 g to 0.0847 g) and E_{miss} (+17%, from 2.997 m to 3.507 m) across all three ablations, while E_t changes by only 3.5%. This non-uniform degradation demonstrates

that the backbone contributes primarily to control-relevant feature extraction and terminal precision rather than engagement duration—matching its theoretical role of resolving the coupled LOS, relative-motion, and time-to-go state features needed for low-lateral-acceleration precision guidance.

Reviewer 2

Comment 2.1

Reviewer comment 2.1

The author lists "introducing a reinforcement learning-based reward function to achieve precise time coordination" as the third core contribution. However, in the Multi-Agent Reinforcement Learning (MARL) field, adding consensus-based penalty terms to the reward function (Equation 64) to guide agents toward consistency is a rather common reward-shaping technique. "Dual clipping with adaptive KL" is a common method used in RL training to prevent gradient explosion, yet the author packages it as a core innovation for solving "aircraft overload saturation".

Response. We sincerely thank the reviewer for this important and insightful comment. We fully agree that consensus-based reward shaping, Dual-Clip PPO, and adaptive KL regularization are established reinforcement-learning techniques. The original manuscript did not distinguish sufficiently between the novelty of a general learning mechanism and its task-specific use in cooperative interception. In particular, the third contribution overstated the novelty of the reward formulation, while several descriptions incorrectly associated Dual-Clip and adaptive KL regularization directly with aircraft acceleration saturation. We acknowledge that these statements were imprecise and could misrepresent the technical contribution of the work.

We have therefore revised the manuscript to make two points explicit. First, the contribution of the reward design lies in encoding the time-coordination requirement of cooperative interception through the groupwise time-to-go penalty, rather than in claiming consensus-based reward shaping itself as a new MARL technique. Second, Dual-Clip PPO and adaptive KL regularization are now consistently described as training-stage mechanisms for stabilizing policy updates, rather than as direct constraints on aircraft acceleration commands. The seven corresponding revisions are listed below. In each white box, deleted text is shown in red and added text is shown in blue, consistent with the revised manuscript.

(1) Reframing the third contribution.

Location of the revision. Section I, in the third item of the contribution list. We removed the statement that presented the introduction of a reinforcement-learning reward function itself as a core contribution. The revised statement instead identifies the task-specific role of the groupwise time-to-go penalty.

A cooperative interception method augmented with a reinforcement learning-based reward function is introduced, enabling precise temporal coordination. This strategy enhances interception success rates against large-scale maneuvering swarm targets by addressing dynamic coordination challenges.

The time-coordination requirement in cooperative interception is incorporated into ART-MAPPO through a groupwise time-to-go penalty. This task-specific formulation converts terminal timing constraints into stepwise learning feedback, enabling coordinated interceptor arrival under target maneuvers.

(2) Revising the overall positioning of Dual-Clip and adaptive KL.

Location of the revision. Section IV, in the opening paragraph of "ART-MAPPO: Attention-Residual and Trust-Aware MAPPO for Cooperative Swarm Interception." We removed Dual-Clip and adaptive KL from the list of task-oriented modules and deleted the claim that they directly constrain guidance-command variations against actuator saturation. Their role is now stated as policy-update stabilization during training.

> Building on the distributed consensus-based IDBO target assignment, we propose the Attention-Residual and Trust-aware MAPPO (ART-MAPPO) algorithm for the cooperative guidance of defending UAV swarms under the Centralized Training with Decentralized Execution (CTDE) paradigm. ART-MAPPO augments the standard MAPPO framework with three synergistic modules:two task-oriented modules: a kinematics-aware attention-residual network for extracting cross-coupled engagement features,and a trust-modulated exploration mechanism for discovering cooperative tactics within the safe flight envelope, and a dual-clip PPO with adaptive KL penalty for constraining guidance command variations against actuator saturation. Dual-Clip PPO and adaptive KL regularization are used to stabilize policy updates during training.

(3) Correcting the stated consequence of large importance-ratio variations.

Location of the revision. Section IV, at the beginning of "Robust Optimization via Dual-Clip and Adaptive KL." The previous text linked importance-ratio variations directly to oscillatory terminal acceleration commands. It has been replaced by the corresponding optimization-level consequence: increased variance of policy updates.

> Aggressive exploration inherently induces can induce substantial variations in the importance-sampling ratio $r_t(\boldsymbol{\vartheta}) = \pi_{\boldsymbol{\vartheta}}(\mathbf{a}_t \mid o_t)/\pi_{\boldsymbol{\vartheta}_{\text{old}}}(\mathbf{a}_t \mid o_t)$, leading to oscillatory terminal lateral-acceleration commands n_{y_i} that can severely compromise flight stability increasing the variance of policy updates.

(4) Correcting the explanation of the standard clipped surrogate.

Location of the revision. Section IV-C (*Robust Optimization via Dual-Clip and Adaptive KL*), immediately after the standard clipped-surrogate objective in Eq. (48) and before the Dual-Clip objective in Eq. (49). We replaced the asserted acceleration-command effect with the actual mini-batch optimization issue caused by large-ratio, negative-advantage samples.

> When $\hat{A}_t < 0$, the standard clip permits $r_t(\boldsymbol{\vartheta}) \to \infty$, causing catastrophic over-correction of acceleration commands allowing large-ratio negative-advantage samples to exert disproportionate influence on a mini-batch update. The Dual-Clip objective addresses this by introducing a floor $c > 1 + \epsilon$ applied only when $\hat{A}_t < 0$.

(5) Adding an intuitive explanation of the Dual-Clip objective.

Location of the revision. Section IV-C (*Robust Optimization via Dual-Clip and Adaptive KL*), immediately after the Dual-Clip objective in Eq. (49) and before the effective probability-ratio constraint in Eq. (50). We added an objective-level explanation that identifies the affected samples and clarifies the role of the additional maximum operation.

For a negative-advantage sample, an exceptionally large probability ratio can otherwise dominate the mini-batch update. The additional maximum limits this negative surrogate contribution, while positive-advantage samples retain the standard clipped objective.

(6) Correcting the interpretation of adaptive KL regularization.

Location of the revision. Section IV-C (*Robust Optimization via Dual-Clip and Adaptive KL*), in the paragraph immediately following the adaptive-KL coefficient update in Eq. (52) and before the value-loss definition in Eq. (53). We removed the direct association between KL divergence and acceleration-limit exceedance and now describe the mechanism in terms of policy-distribution shift and regularization strength.

When $\hat{D}_{\mathrm{KL}} > 2\delta_{\mathrm{targ}}$, the acceleration commands have shifted too aggressively between updates, risking acceleration-limit exceedancepolicy distribution has shifted excessively between updates; the $1.5\times$ increase dampens subsequent updatesstrengthens regularization in subsequent updates. When $\hat{D}_{\mathrm{KL}} < 0.5\delta_{\mathrm{targ}}$, the $0.9\times$ decay permits finer trajectory refinementfurther policy refinement.

(7) Clarifying the roles of the terms in the overall training loss.

Location of the revision. Section IV-C (*Robust Optimization via Dual-Clip and Adaptive KL*), in the paragraph immediately following the overall ART-MAPPO loss in Eq. (54) and before the subsequent discussion of its computational cost. We added a concise explanation distinguishing the roles of the actor objective, KL and entropy regularization, and centralized critic loss.

Minimizing this loss updates the actor through the Dual-Clip objective, while the KL and entropy terms regulate policy change and exploration. The value-loss term trains the centralized critic.

Comment 2.2

Reviewer comment 2.2

The module schematic diagram on the left side of Fig. 5 lacks clarity in its presentation. $h^{(0)}$ appears at multiple locations, and the flow from $h^{(\ell)}$ to $h^{(\ell+1)}$ is unclear.

Response. We sincerely thank the reviewer for pointing out this presentation problem. We agree that the repeated appearance of $\mathbf{h}^{(0)}$ and the unclear transition from $\mathbf{h}^{(\ell)}$ to $\mathbf{h}^{(\ell+1)}$ reduced the clarity of the original schematic. We have carefully revised the module schematic diagram on the left side of Fig. 5 to address these issues. The revised figure is reproduced below.

Location of the revision. Section IV, Fig. 5.

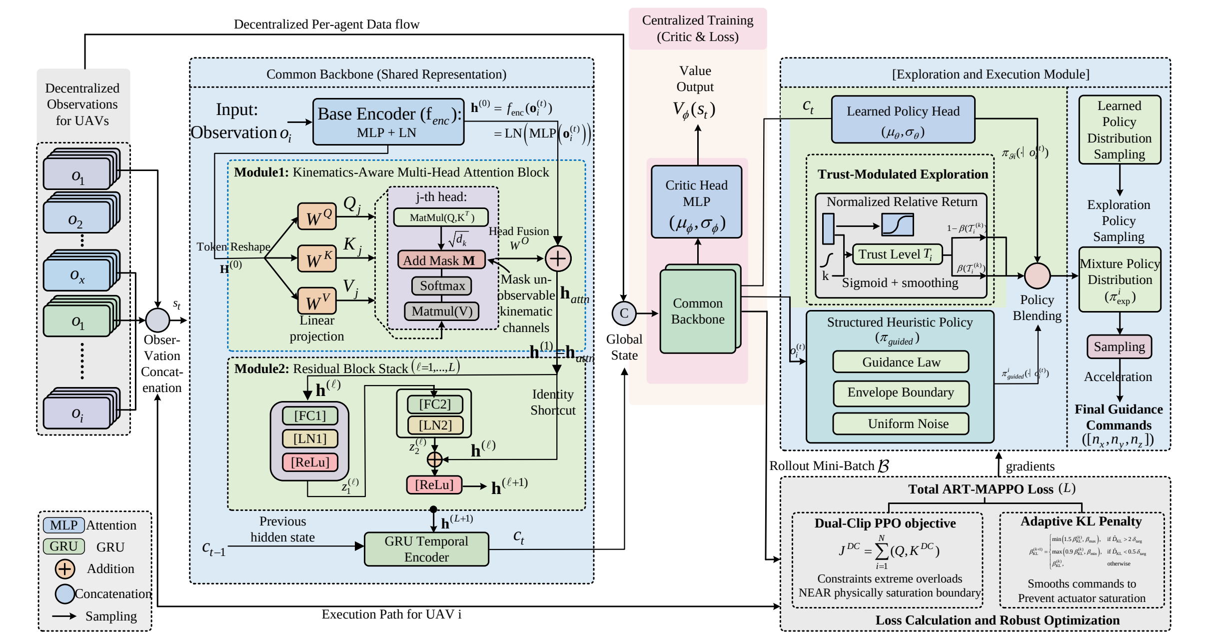

Fig. 5. Overall workflow of the ART-MAPPO algorithm for cooperative swarm interception, showing the interaction among the attention-residual network, trust-modulated exploration, dual-clip adaptive KL optimization, and the three-dimensional acceleration-command output.

Comment 2.3

Reviewer comment 2.3

Section IV.B contains a logical paradox: the text describes a monotonic incremental process, while Equation (42) implements a convergent smoothing process. This divergence between the mathematical definition and physical intuition renders the mechanism logically inconsistent in maintaining high-trust stability.

Response. We sincerely thank the reviewer for identifying this logical inconsistency. We fully agree with the reviewer and acknowledge that the physical interpretation provided in the original manuscript was inaccurate.

Equation (42) is the intended trust-update definition and has been retained unchanged. We deleted the inaccurate sentence and replaced it with an explanation consistent with the smoothing process implemented by the equation. The revised text clarifies how the bounded performance reference is formed, how the previous trust level and current reference are combined, and how smoothing attenuates episode-to-episode fluctuations.

Location of the revision. Section IV-B (*Trust-Modulated Tactical Exploration*), immediately following Eq. (42) and before the paragraph *2) Blended Exploration Policy*. Equation (42) remains unchanged; the original interpretation was deleted and replaced as follows:

$$\mathcal{T}_i^{(k+1)} = (1-\alpha_T)\mathcal{T}_i^{(k)} + \alpha_T \sigma\left(\tau_T \tilde{R}_i^{(k)}\right). \tag{42}$$

where $\sigma(x) = 1/(1+e^{-x})$, $\alpha_T \in (0,1)$ is the trust update rate, and $\tau_T > 0$ is the temperature parameter controlling the sensitivity of trust to performance deviations.
When $R_i^{(k)} > \mu_R$, the sigmoid output exceeds 0.5, driving $\mathcal{T}_i$ upward and reducing exploration; conversely, below-average performance lowers $\mathcal{T}_i$ and amplifies exploration.
The sigmoid term maps the normalized relative return $\tilde{R}_i^{(k)}$ to a bounded performance reference in $(0,1)$. Equation (42) combines the previous trust level and the current performance reference with the weights $1-\alpha_T$ and α_T, respectively, yielding a smoothed trust estimate. The smoothing process tracks changes in relative performance while attenuating episode-to-episode return fluctuations. Sustained above-average returns keep the performance reference above 0.5 and bias the smoothed trust estimate toward a correspondingly higher level, thereby reducing the probability assigned to the guided exploratory policy; sustained below-average returns produce the opposite allocation.

The revised physical interpretation is now consistent with the smoothing process defined by Eq. (42). In particular, high-trust stability results from sustained high performance references together with the attenuation of episode-level fluctuations, rather than from an unconditionally increasing trust value.

Comment 2.4

Reviewer comment 2.4

The simulation environment is rather simplistic, and apart from UAV dynamics, there is no other environmental modeling, which does not align with the high-fidelity simulation environment claimed in the paper.

Response. We sincerely thank the reviewer for this important observation. We agree that the original planar and idealized simulation environment did not sufficiently support the claimed level of fidelity. In response, we have substantially revised both the engagement model and the experimental environment. The engagement is now formulated in three-dimensional space, while communication delay, sensing uncertainty, and communication packet loss are explicitly incorporated into the observation process. In addition, the complete target-assignment and cooperative-guidance framework is implemented on a hardware-in-the-loop (HIL) platform comprising multiple independent embedded computing nodes and a physical communication network. All experiments reported in the revised manuscript are based on this updated three-dimensional HIL environment and the stated communication and sensing conditions. The specific revisions are detailed below.

(1) Extension from planar motion to three-dimensional engagement dynamics.

Location of the revision. Section II-A (*Problem Modeling*), Eqs. (1)–(3) and (5). The original planar kinematic, relative-motion, acceleration, and ZEM definitions were replaced by three-dimensional formulations.

The object of study in this paper is the fixed-wing UAV, and the interception problem is reduced to a two-dimensional horizontal plane formulated in three-dimensional space, as shown in Fig.2.

$$
\begin{cases} \dot{x} = V\cos\theta\cos\gamma, \\ \dot{y} = V\cos\theta\sin\gamma, \\ \dot{z} = V\sin\theta, \end{cases} \tag{1}
$$

(x, y, z) are the three-dimensional position coordinates, V is the velocity magnitude, γ is the heading angle, and θ is the flight-path angle.

Let $\mathbf{p}_A, \mathbf{p}_D \in \mathbb{R}^3$ and $\mathbf{v}_A, \mathbf{v}_D \in \mathbb{R}^3$ denote the attacker and defender states. Their three-dimensional relative geometry is

$$
\begin{aligned} \mathbf{r}_{AD} &= \mathbf{p}_A - \mathbf{p}_D, \qquad r = \|\mathbf{r}_{AD}\|_2, \\ \boldsymbol{\ell}_{AD} &= \mathbf{r}_{AD}/r, \qquad \dot{r} = \boldsymbol{\ell}_{AD}^{\top}(\mathbf{v}_A - \mathbf{v}_D). \end{aligned} \tag{2}
$$

where r and $\boldsymbol{\ell}_{AD}$ denote the three-dimensional relative distance and LOS unit vector, respectively. As illustrated in Fig.2, q_{ij}^{γ} and q_{ij}^{θ} are the corresponding LOS azimuth and elevation angles.

$$
\begin{cases} \dot{V} = n_x g, \\ \dot{\gamma} = \dfrac{n_y g}{V\cos\theta}, \\ \dot{\theta} = \dfrac{n_z g}{V}, \end{cases} \tag{3}
$$

where n_x, n_y, and n_z denote the axial, lateral, and normal accelerations, and g denotes gravitational acceleration.

$$
\text{ZEM} = \left\| \mathbf{r}_{AD} + (\mathbf{v}_A - \mathbf{v}_D)t_{go} \right\|_2. \tag{5}
$$

(2) Incorporation of communication delay and sensing uncertainty.

Location of the revision. Section V-A (*Experimental Setup and Implementation Details*), in the newly added paragraph *Communication Delay and Sensing Uncertainty*, including Eq. (71). The observations are now constructed from delayed and noisy Cartesian states, and communication packet loss is included in the information exchange process.

1) Communication Delay and Sensing Uncertainty: Communication and sensing nonidealities are incorporated into the three-dimensional simulation. The position and velocity measurements used to construct the defender observations are delayed and corrupted by independent zero-mean Gaussian noise,

$$
\widehat{\mathbf{p}}(t) = \mathbf{p}(t - \tau_s) + \boldsymbol{\eta}_p, \qquad \widehat{\mathbf{v}}(t) = \mathbf{v}(t - \tau_s) + \boldsymbol{\eta}_v. \tag{71}
$$

where $\boldsymbol{\eta}_p \sim \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I}_3)$ and $\boldsymbol{\eta}_v \sim \mathcal{N}(\mathbf{0}, \sigma_v^2 \mathbf{I}_3)$. The nominal settings are $\tau_s = 50$ ms, $\sigma_p = 1$ m, and $\sigma_v = 0.3$ m/s. All derived observation variables, including the LOS geometry, closing speed, and time-to-go, are recomputed from the delayed noisy Cartesian states. The communication process is subject to a 50 ms communication delay and a 1% packet-loss rate, with the latest valid state packet retained when a packet is lost.

(3) HIL Platform.

Location of the revision. Section V-A, in the newly added paragraph *HIL Platform*, immediately following Eq. (71), together with the platform architecture shown in Fig. 7. The policies and target-assignment modules of five defenders are executed on independent NVIDIA Jetson NX computers, while the engagement dynamics and remaining agents are executed on the simulation server.

2) HIL Platform: As shown in Fig.7, the three-dimensional interception experiment is further implemented on a HIL platform. The policies of defenders D_0–D_4 are deployed on five independent NVIDIA Jetson NX computers, with one defender assigned to each node. The corresponding target-assignment algorithm is also executed on the Jetson NX nodes. The engagement dynamics, attacker cluster, and software policies of the remaining defenders D_5–D_n are executed on the simulation server. The server and Jetson NX nodes exchange observations, actions, and target-assignment results through a physical TCP Ethernet network under the communication conditions described above.

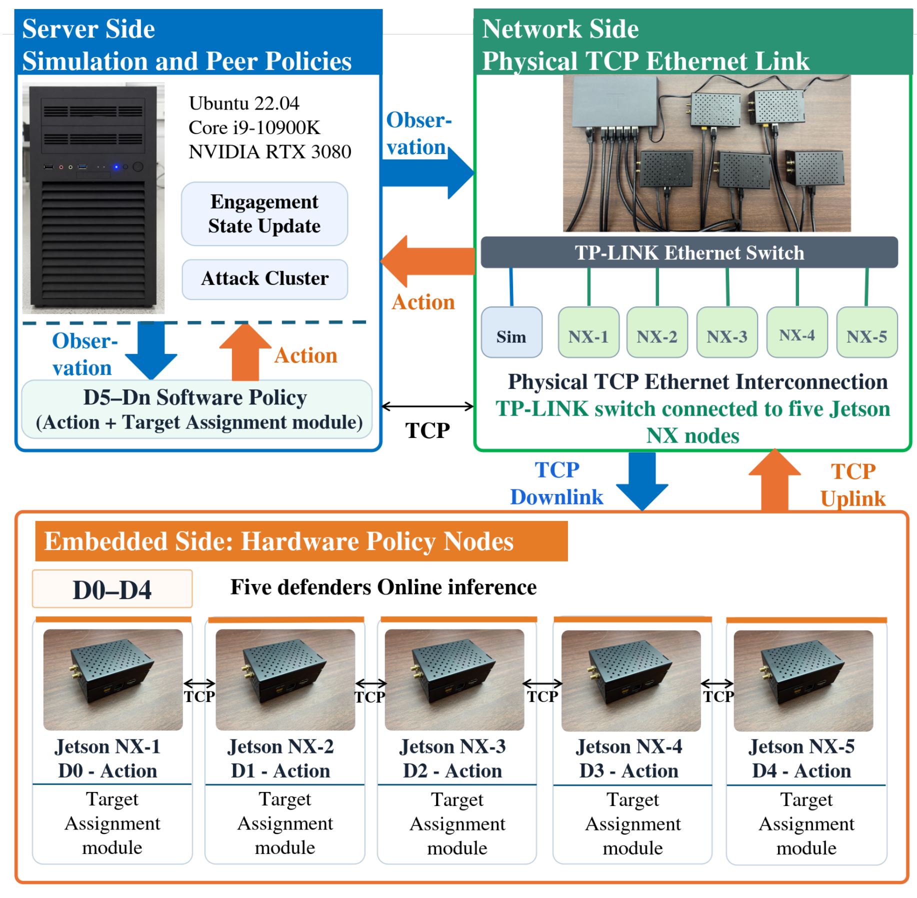

Fig. 7. HIL platform.

(4) Use of the revised environment in the reported experiments.

Location of the revision. Section V-D (*Cooperative Interception against Maneuvering Targets*), at the beginning of the subsection. We clarified that the complete target-assignment and cooperative-guidance process is evaluated using the three-dimensional dynamics and the communication and sensing conditions specified above.

We evaluate the complete target-assignment and terminal cooperative-guidance process against two maneuvering scenarios, each benchmarked against classical PN. The experiments are conducted on the HIL platform with communication delays, sensing uncertainty and three-dimensional dynamics. The target-assignment and cooperative-guidance modules are executed sequentially to provide end-to-end validation of the complete interception process.

Comment 2.5

Reviewer comment 2.5

In Section V.C, the authors state that each episode has at most 25 steps during training, while in Section V.A the simulation step size is set to 0.05s. This implies a training period of 1.25s, which is impossible to learn any interception strategies given an engagement distance of 1500 meters and a maximum speed of 40 m/s. By the way, the settings "$D_1 = 100$ m, $D_2 = 1500$ m, and $D_3 = 1600$ m" are inconsistent with the illustrations in Fig. 6.

Response. We sincerely thank the reviewer for carefully identifying these two inconsistencies. We fully agree with the reviewer. Both resulted from incorrect parameter values reported in the original manuscript.

First, the maximum episode horizon was incorrectly written as 25 steps. The training implementation and all reported results use a maximum horizon of 2500 steps. We have corrected this typographical error in the revised manuscript.

Second, the values assigned to D_1, D_2, and D_3 were inadvertently permuted in the original description. In the actual experimental configuration, $D_1 = 1500$ m and $D_2 = 1600$ m define the inner and outer radii of the annular region in which the attacking UAVs are initialized, whereas $D_3 = 100$ m defines the radius of the defender initialization region. These corrected values are consistent with Fig. 6 and the initial state distributions in Table II.

The corresponding revisions are provided below.

(1) Correction of the maximum episode horizon.

Location of the revision. Section V-C (*RL Model Training and Convergence Analysis*), in the first paragraph of the subsection. The incorrectly reported horizon of 25 steps was corrected to 2500 steps.

To evaluate the learning efficiency and robustness of the proposed ART-MAPPO, comparative training experiments were conducted against standard MAPPO, IPPO, IA2C, and IQL. All models were trained over 10,000 episodes with a maximum horizon of ~~25~~2500 steps per episode.

(2) Correction of the engagement-region parameters.

Location of the revision. Section V-A (*Experimental Setup and Implementation Details*), in the paragraph immediately following Fig. 6 and before Table II. The values of D_1, D_2, and D_3 were corrected to match the engagement geometry, and the horizontal-position coordinates were clarified.

A representative scenario of 20 D_{UAV} versus 8 A_{UAV} is considered, with $D_1 = 1001500\,\mathrm{m}$, $D_2 = 15001600\,\mathrm{m}$, and $D_3 = 1600100\,\mathrm{m}$. The initial horizontal position is specified by the radial coordinate r and polar angle φ about the defended-area center.

Comment 2.6

Reviewer comment 2.6

There is a lack of end-to-end experimental verification covering the complete workflow of "task allocation and cooperative guidance". Furthermore, ablation studies are not provided to demonstrate the effectiveness of the proposed innovations.

Response. We sincerely thank the reviewer for this important and constructive comment. We agree that the original experimental results did not sufficiently demonstrate the complete workflow from target assignment to cooperative guidance, nor did they isolate the contribution of the principal components of ART-MAPPO.

We have addressed these two issues separately. First, after extending the experimental environment to three-dimensional engagement dynamics with communication delay and sensing uncertainty, as detailed in our response to Comment 2.4, we reran the simulation experiments as end-to-end evaluations. In each trial, the defending swarm first performs target assignment and then executes cooperative guidance according to the resulting interceptor–target groups. The terminal results therefore reflect the complete sequential workflow rather than an independently initialized guidance stage. The two end-to-end engagement cases are presented in Section V-D, and their statistical performance over 1,000 independent trials per case is reported in Section V-F.

Second, we added a dedicated component-ablation subsection. Three one-at-a-time controls remove the trust-aware mechanism, GRU temporal encoder, and attention–residual backbone, respectively. The complete model and all ablated variants use the same reward, dynamics, optimizer, training seed, training budget, and Monte Carlo evaluation conditions. The resulting training and interception metrics identify the distinct contribution of each component.

The corresponding revisions are provided below.

(1) End-to-end verification of target assignment and cooperative guidance.

Location of the revision. Section V-D (*Cooperative Interception against Maneuvering Targets*), including the introductory paragraph and the descriptions of Cases 1 and 2. Section V-F (*Monte Carlo Statistical Analysis and Robustness*) further reports 1,000 end-to-end Monte Carlo trials for each case.

We evaluate the complete target-assignment and terminal cooperative-guidance process against two maneuvering scenarios, each benchmarked against classical PN. The experiments are conducted on the HIL platform with communication delays, sensing uncertainty and three-dimensional dynamics. The target-assignment and cooperative-guidance modules are executed sequentially to provide end-to-end validation of the complete interception process.

Case 1: Interception against Evasive Maneuvering Targets
In this scenario, the attacking UAVs (A_{UAV}) initially execute a constant-g evasive maneuver and transition to proportional navigation (PN) guidance after 15 s. This case provides an end-to-end validation in which the defending swarm performs target assignment and then executes

cooperative guidance against these highly dynamic threats.

Case 2: Interception against Continuous Weaving Targets
This scenario evaluates the robustness of the proposed framework against the sinusoidal penetration maneuver [52], which represents a highly agile and continuous weaving threat. Such maneuvers induce high-frequency oscillations in the line-of-sight (LOS) rate, severely challenging both the guidance stability and the temporal coordination of the defending swarm. This case provides an end-to-end validation in which target assignment is followed by cooperative guidance against the three-dimensional weaving threats.

The end-to-end Monte Carlo evaluation in Section V-F comprises 1,000 independent trials for each case. The complete proposed framework successfully intercepts all targets in 987 trials in Case 1 and 971 trials in Case 2, corresponding to interception success rates of 98.7% and 97.1%, respectively. These results evaluate the combined outcome of target assignment and subsequent cooperative guidance.

(2) Component ablation of ART-MAPPO.

Location of the revision. Section V-E, newly added subsection *Component Ablation Study*, including Table V and the corresponding component-level analysis. The complete added subsection is reproduced below.

Component Ablation Study

To isolate the independent contribution of each proposed component, we construct three one-at-a-time ablation controls alongside the complete ART-MAPPO model. *"w/o trust-aware mechanism"* disables the trust-modulated fusion of the analytical guidance prior and the learned action; *"w/o GRU temporal encoder"* replaces recurrent state encoding with a memoryless feedforward layer; *"w/o attention–residual backbone"* substitutes a parameter-matched feedforward network while retaining the trust and GRU modules. All variants share the same reward, dynamics, optimiser, training seed, and budget, and are evaluated in 1000 frozen-policy Case 1 Monte Carlo trials. Table V summarises the results. Critically, the three controls exhibit *module-specific* rather than uniformly ordered degradation, providing direct evidence that each component carries a distinct and non-redundant role.

Table V. Component ablation of ART-MAPPO: 1000-trial Case 1 Monte Carlo evaluation.

Variant	Training AUC (10^4) ↑	Final reward ↑	ISR (%) ↑	$E_{co\text{-}time}$ (s) ↓	E_n (g) ↓	E_t (s) ↓	E_{miss} (m) ↓
Full ART-MAPPO	7.354	8.958	**98.7**	0.0505	0.0718	28.843	2.997
w/o trust-aware mechanism	6.037	8.683	97.2	0.0545	0.0739	29.132	3.087
w/o GRU temporal encoder	6.388	8.134	95.8	0.0732	0.0782	30.574	3.297
w/o attention–residual backbone	6.651	8.353	96.1	0.0606	0.0847	29.853	3.507

(i) Trust-aware mechanism. Removing trust produces the largest training-AUC reduction among the three controls (−17.9%, from 7.354×10^4 to 6.037×10^4), yet the final reward falls by only 3.1% and ISR drops by 1.5 pp. The pronounced gap between cumulative-AUC loss and end-of-training reward confirms that the trust-aware mechanism primarily accelerates learning efficiency throughout training, not merely raises the asymptotic policy ceiling—consistent with its design role of dynamically regulating exploration–exploitation balance.

(ii) GRU temporal encoder. The GRU removal yields the lowest final reward (-9.2%) and the largest ISR degradation (-2.9 pp) among all controls. Most distinctively, $E_{co\text{-}time}$ increases by 45% (from 0.0505 s to 0.0732 s), far exceeding the 6% increase in E_t. Without temporal history, the policy can no longer reliably track target manoeuvres or reconcile interceptor time-to-go estimates, leading to simultaneous degradation in both final policy quality and cooperative timing precision.

(iii) Attention–residual backbone. Removing the attentive backbone yields the largest increases in E_n (+18%, from 0.0718 g to 0.0847 g) and E_{miss} (+17%, from 2.997 m to 3.507 m) across all three ablations, while E_t changes by only 3.5%. This non-uniform degradation demonstrates that the backbone contributes primarily to control-relevant feature extraction and terminal precision rather than engagement duration—matching its theoretical role of resolving the coupled LOS, relative-motion, and time-to-go state features needed for low-lateral-acceleration precision guidance.

Reviewer 3

Comment 3.1

Reviewer comment 3.1

The author is encouraged to further justify the distributed consensus-based IDBO formulation by clarifying how adaptive rolling, dancing, breeding and stealing coefficients influence convergence stability in interception scenarios.

Response. We sincerely thank the reviewer for this valuable suggestion. We have added a detailed discussion immediately following the rolling, dancing, breeding, and stealing operators. The revised text defines a common linear decay schedule for their adaptive coefficients and explains its effect over the IDBO iterations. During the early iterations, relatively large coefficients preserve the exploratory capability needed to search among competing interceptor–target assignments. As the iterations proceed, the rolling and dancing perturbations diminish, while the magnitudes of the breeding- and stealing-directed preference updates are progressively reduced. This transition suppresses late-stage assignment oscillations and allows the working solution to settle through deterministic, advantage-weighted local refinement and subsequent binarization. The corresponding revision is reproduced below.

Location of the revision. Section III-B (*Distributed Consensus-Based IDBO Framework*), immediately following the rolling, dancing, breeding, and stealing operators in Eqs. (16)–(19), and before the advantage-adaptive thresholding operation in Eq. (20). The following explanation has been added:

> The coefficients $\alpha, \beta, \gamma_{\mathrm{IDBO}}, \delta, \delta', \eta_{\mathrm{IDBO}}$, together with the dancing-noise scale σ_d, follow the common linear schedule $c_t = c_0(1 - t/T)$, where T denotes the local IDBO search horizon. During each local search, the candidate preferences are projected onto $[0,1]^N$, and the best-so-far candidate is retained unless a strictly better candidate is obtained. The coefficient schedule progressively attenuates both the directed update terms and the stochastic perturbations. In particular, the rolling perturbation $\beta \boldsymbol{\xi}_i \mathrm{e}^{-t/T}$ and the dancing perturbation $\mathbf{n}_i \sim \mathcal{N}(0, \sigma_d^2 \mathbf{I})$ vanish as t approaches T, suppressing late-stage fluctuations in the preference population. The schedule therefore satisfies the vanishing-gain condition $c_t \to 0$ used in Robbins–Monro-type stochastic approximation [51]. At $c_T = 0$, the four update operators reduce to the identity map, while elitist retention prevents the incumbent from being replaced by an inferior candidate. Consequently, once no strictly improving candidate is generated, the retained solution remains invariant and constitutes a stable fixed point of the implemented local search.

Comment 3.2

Reviewer comment 3.2

Also, a more rigorous complexity and scalability analysis is needed for the Improved Dung Beetle Optimization algorithm under cooperative deflection environments where communication delays influence distributed consensus efficiency.

Response. We sincerely thank the reviewer for this important and constructive suggestion. We agree that the computational scalability of IDBO and the influence of communication delay on distributed consensus should be analyzed separately and supported by quantitative results. Accordingly, we have extended both the theoretical analysis in Section III-B and the experimental evaluation in Section V-B.

The revised theoretical analysis distinguishes the local computational cost from the communication-side consensus cost. Based on the per-UAV per-iteration complexity in Eq. (27), we clarify the effects of jointly increasing the interceptor and target populations. We also added two quantitative evaluations. The first measures the CPU time of IDBO over assignment scales ranging from 10×4 to 160×64. The second evaluates distributed consensus under communication delays from 50 to 450 ms using the nominal 20×8 configuration.

The corresponding revisions are provided below.

(1) Complexity and communication-scalability analysis.

Location of the revision. Section III-B (*Distributed Consensus-Based IDBO Framework*), immediately after the per-UAV complexity model in Eq. (27). The following analysis has been added:

> As M and N increase together, the overall computational workload also increases, as quantified by the scalability results in Section V-B. Communication-side scalability is governed by the convergence efficiency of the distributed consensus process. Once the bids have stabilized, the winner lists for all N targets are exchanged together and propagate through a connected graph of diameter D within at most D successful communication rounds. The corresponding communication time scales as $O(DT_{\text{comm}})$ under bounded per-hop delay T_{comm}, while the per-round communication processing cost is $O(N|\mathcal{N}_i|)$. The experiments in Section V-B show that increasing communication delay increases both the mean communication steps and the consensus time, while all tested cases reach common top-$L_{\max}$ winner lists.

(2) Computational-scalability experiment.

Location of the revision. Section V-B (*Target Assignment Performance and Spatial Topology*), following the target-assignment topology results. We added a scale-sweep experiment using five interceptor–target configurations:

> To further examine the computational scalability and delay-dependent consensus efficiency of IDBO, the nominal 20×8 configuration is used as the baseline, and the scale sweep considers 10×4, 20×8, 40×16, 80×32, and 160×64. Under a fixed optimization budget, the mean CPU times are 0.031 s, 0.048 s, and 1.948 s for the representative 10×4, 20×8, and 160×64 configurations, respectively. The observed runtime scaling is superlinear over the evaluated configurations when the interceptor and target populations are increased concurrently. These CPU times quantify the computational cost of IDBO and exclude distributed communication latency.

(3) Delay-dependent consensus experiment.

Location of the revision. Section V-B, immediately after the computational-scalability analysis. We added a communication-delay sweep for the nominal 20×8 assignment. The corresponding table and analysis are reproduced below:

> Table IV. Delay-dependent consensus performance of IDBO for the nominal 20×8 assignment.

Communication delay (ms)	Mean communication steps	Mean consensus time (s)	Consensus success (%)
50	10.6	0.53	100
100	19.2	0.96	100
150	27.8	1.39	100
250	45.0	2.25	100
450	79.4	3.97	100

Table IV reports the distributed consensus performance at the nominal assignment scale. The 50-ms case corresponds to the nominal communication condition, while the remaining cases provide an extended-delay evaluation. The communication-step count is measured on the 50-ms communication time grid. As the communication delay increased from 50 to 450 ms, the mean number of communication steps required to reach consensus increased from 10.6 to 79.4, and the corresponding mean consensus time increased from 0.53 to 3.97 s. Nevertheless, the consensus success rate remained 100% under all tested delays. These results show that bounded communication delay reduces distributed consensus efficiency but does not prevent convergence to a common conflict-free assignment under the tested conditions.

Comment 3.3

Reviewer comment 3.3

Then, it would be helpful to add an ablation investigation demonstrating the independent contribution of the trust-aware mechanism, GRU temporal encoder, and attention-residual backbone within ART-MAPPO architecture.

Response. We sincerely thank the reviewer for this valuable suggestion. We agree that the independent contributions of the principal ART-MAPPO components should be established through controlled ablation rather than inferred only from the comparison with other reinforcement-learning algorithms.

Accordingly, we added a dedicated component-ablation study in Section V-E. Three one-at-a-time variants remove the trust-aware mechanism, GRU temporal encoder, and attention–residual backbone, respectively. To ensure a controlled comparison, all variants use the same reward function, engagement dynamics, optimizer, training seed, and training budget. Each trained policy is subsequently evaluated in 1,000 frozen-policy Monte Carlo trials under Case 1.

The corresponding revisions are provided below.

Location of the revision. Section V-E, newly added subsection *Component Ablation Study*, including Table V and the corresponding component-level analysis. The complete added subsection is reproduced below.

Component Ablation Study

To isolate the independent contribution of each proposed component, we construct three one-at-a-time ablation controls alongside the complete ART-MAPPO model. *"w/o trust-aware mechanism"* disables the trust-modulated fusion of the analytical guidance prior and the learned action; *"w/o GRU temporal encoder"* replaces recurrent state encoding with a memoryless feedforward layer; *"w/o attention–residual backbone"* substitutes a parameter-matched feedforward network while retaining the trust and GRU modules. All variants share the same reward, dynamics, optimiser, training seed, and budget, and are evaluated in 1000 frozen-policy Case 1 Monte Carlo trials. Table V summarises the results. Critically, the three controls exhibit *module-specific* rather than uniformly ordered degradation, providing direct evidence that each component carries a distinct and non-redundant role.

Table V. Component ablation of ART-MAPPO: 1000-trial Case 1 Monte Carlo evaluation.

Variant	Training AUC (10^4) ↑	Final reward ↑	ISR (%) ↑	$E_{co\text{-}time}$ (s) ↓	E_n (g) ↓	E_t (s) ↓	E_{miss} (m) ↓
Full ART-MAPPO	7.354	8.958	**98.7**	0.0505	0.0718	28.843	2.997
w/o trust-aware mechanism	6.037	8.683	97.2	0.0545	0.0739	29.132	3.087
w/o GRU temporal encoder	6.388	8.134	95.8	0.0732	0.0782	30.574	3.297
w/o attention–residual backbone	6.651	8.353	96.1	0.0606	0.0847	29.853	3.507

(i) Trust-aware mechanism. Removing trust produces the largest training-AUC reduction among the three controls (-17.9%, from 7.354×10^4 to 6.037×10^4), yet the final reward falls by only 3.1% and ISR drops by 1.5 pp. The pronounced gap between cumulative-AUC loss and end-of-training reward confirms that the trust-aware mechanism primarily accelerates learning efficiency throughout training, not merely raises the asymptotic policy ceiling—consistent with its design role of dynamically regulating exploration–exploitation balance.

(ii) GRU temporal encoder. The GRU removal yields the lowest final reward (-9.2%) and the largest ISR degradation (-2.9 pp) among all controls. Most distinctively, $E_{co\text{-}time}$ increases by 45% (from 0.0505 s to 0.0732 s), far exceeding the 6% increase in E_t. Without temporal history, the policy can no longer reliably track target manoeuvres or reconcile interceptor time-to-go estimates, leading to simultaneous degradation in both final policy quality and cooperative timing precision.

(iii) Attention–residual backbone. Removing the attentive backbone yields the largest increases in E_n (+18%, from 0.0718 g to 0.0847 g) and E_{miss} (+17%, from 2.997 m to 3.507 m) across all three ablations, while E_t changes by only 3.5%. This non-uniform degradation demonstrates that the backbone contributes primarily to control-relevant feature extraction and terminal precision rather than engagement duration—matching its theoretical role of resolving the coupled LOS, relative-motion, and time-to-go state features needed for low-lateral-acceleration precision guidance.

Comment 3.4

Reviewer comment 3.4

Correspondingly, including a comparative theoretical discussion between the proposed ART-MAPPO framework and conventional MAPPO variants regarding convergence guarantees, exploration–exploitation balance, and policy robustness would significantly strengthen the research technically.

Response. We thank the reviewer for this valuable suggestion. We have added a new highlighted subsection entitled "Mechanistic Comparison with Conventional MAPPO" in Section IV, which compares ART-MAPPO with conventional MAPPO along the three aspects identified by the reviewer.

The corresponding revisions are provided below.

Location of the revision. Section IV-D, newly added subsection *Mechanistic Comparison with Conventional MAPPO*, immediately following *Robust Optimization via Dual-Clip and Adaptive KL*. The complete subsection is reproduced below.

> *Mechanistic Comparison with Conventional MAPPO*
>
> *1) Convergence analysis.* Conventional MAPPO primarily relies on centralized value estimation, GAE, and a fixed clipped surrogate to support policy learning. In ART-MAPPO, the attention–residual encoder preserves the coupled kinematic information and maintains a direct gradient path, while the GRU incorporates maneuver history into the policy representation. This reduces the sensitivity of policy learning to instantaneous state variations. The smoothed trust variable changes the guided-exploration weight gradually according to relative return, allowing the sampling policy to shift steadily toward exploitation as effective cooperative behavior develops. During parameter optimization, Dual-Clip limits anomalous negative surrogate contributions, and adaptive KL regularization adjusts the penalty according to the observed policy deviation. The resulting learning process exhibits lower sampling and update variability, promoting a more stable approach to the steady training regime than conventional MAPPO.
>
> *2) Exploration–exploitation balance.* In the conventional MAPPO baseline adopted in this study, stochastic action sampling and a fixed-coefficient entropy bonus provide an exploration incentive that is not explicitly conditioned on agent-specific performance. ART-MAPPO retains entropy regularization and additionally introduces the trust-modulated mechanism in (42)–(43). For a fixed normalized return, the trust recursion is a contraction toward a performance-dependent target; during training, it acts as a bounded, exponentially smoothed tracker of relative agent performance. Agents that persistently achieve above-average returns are gradually assigned greater weight to their learned policies, whereas below-average agents retain a larger task-informed policy component. ART-MAPPO therefore adapts the intensity and structure of exploration across agents, rather than relying solely on a uniform entropy incentive.
>
> *3) Policy robustness.* The fixed clipping mechanism of conventional MAPPO cannot adjust its regularization strength according to the policy deviation realized during training. In ART-MAPPO, the adaptive KL mechanism strengthens the penalty when the empirical KL divergence is excessive and relaxes it when the divergence remains below the target range. Combined with Dual-Clip, this feedback reduces the sensitivity of policy optimization to abnormal mini-batches and abrupt inter-update changes.

Comment 3.5

> **Reviewer comment 3.5**
>
> *Meanwhile, it is important to provide sensitivity analysis for weighting coefficients in the block probability model and reward formulation because cooperative interception performance varies under different tactical priorities.*

Response. We sincerely thank the reviewer for this important suggestion. We agree that the weighting coefficients in both the target-assignment probability model and the ART-MAPPO reward function represent different tactical priorities, and that their effects should be demonstrated through end-to-end interception performance rather than only through their mathematical definitions.

We have therefore added two sensitivity studies in Section V-G. First, the IDBO assignment-probability coefficients are varied to examine the tradeoff between interception reachability and three-dimensional engagement geometry. For every coefficient setting, 1,000 end-to-end Monte Carlo trials are conducted in Case 1. In addition to the assignment-stage ZEM and angular-error quantities, the downstream temporal-coordination error, terminal lateral acceleration, interception time, and miss distance are reported.

Second, the five ART-MAPPO reward weights are examined through a one-at-a-time perturbation. Each weight is scaled from 50% to 150% of its nominal value while the remaining weights are held fixed. The resulting cooperative-guidance metrics quantify the tradeoffs among reliability, terminal accuracy, temporal coordination, interception efficiency, and control effort.

The complete additions are reproduced below.

(1) IDBO assignment-weight sensitivity.

Location of the revision. Section V-G (*Parameter Sensitivity Analysis*), in the newly added paragraph *1) IDBO Assignment-Weight Sensitivity* and Table VI.

1) IDBO Assignment-Weight Sensitivity. The coefficients $w_{1,\mathrm{IDBO}}$ and $w_{2,\mathrm{IDBO}}$ represent the relative emphasis on interception reachability and three-dimensional engagement geometry, respectively. The sensitivity analysis is conducted in Case 1 using 1,000 end-to-end Monte Carlo trials for each setting. As shown in Table VI, larger $w_{1,\mathrm{IDBO}}$ tends to reduce $E_{co\text{-}time}$ because reachability-oriented assignments require less arrival-time adjustment, although the resulting engagement geometry may demand greater terminal acceleration and prolong the interception. Increasing $w_{2,\mathrm{IDBO}}$ improves the approach geometry and reduces the maneuvering demand, but the weaker emphasis on reachability increases the difficulty of temporal coordination. The growth of E_{miss} toward either end of the weighting range shows that terminal accuracy deteriorates when either assignment criterion is excessively emphasized.

Table VI. Sensitivity Analysis of the IDBO Assignment-Probability Weighting Coefficients.

$w_{1,\mathrm{IDBO}}$	$w_{2,\mathrm{IDBO}}$	Mean ZEM (m) ↓	Mean 3-D angular error (°) ↓	$E_{co\text{-}time}$ (s) ↓	E_n (g) ↓	E_t (s) ↓	E_{miss} (m) ↓
0.10	0.90	168.0	9.4	0.0718	0.0582	27.980	3.460
0.15	0.85	164.8	9.6	0.0704	0.0576	28.110	3.430
0.20	0.80	161.5	9.8	0.0669	0.0613	28.060	3.310
0.25	0.75	158.4	10.0	0.0643	0.0630	28.260	3.280
0.30	0.70	155.5	10.3	0.0631	0.0624	28.190	3.200
0.35	0.65	152.8	10.7	0.0598	0.0667	28.500	3.140
0.40	0.60	150.4	11.0	0.0584	0.0681	28.610	3.110
0.45	0.55	148.4	11.2	0.0549	0.0675	28.570	3.050
0.50	0.50	147.2	11.4	0.0535	0.0712	28.790	3.030
0.55	**0.45**	**146.4**	**11.5**	**0.0505**	**0.0718**	**28.843**	**2.997**
0.60	0.40	145.4	11.6	0.0491	0.0746	29.020	3.040
0.65	0.35	143.8	11.8	0.0497	0.0760	28.960	3.010
0.70	0.30	141.4	12.1	0.0470	0.0787	29.240	3.100
0.75	0.25	138.3	12.5	0.0465	0.0829	29.180	3.080
0.80	0.20	134.5	12.9	0.0469	0.0815	29.470	3.210
0.85	0.15	130.3	13.3	0.0440	0.0897	29.730	3.270
0.90	0.10	126.0	13.7	0.0448	0.0952	29.690	3.420

(2) ART-MAPPO reward-weight sensitivity.

Location of the revision. Section V-G, in the newly added paragraph *2) ART-MAPPO Reward-Weight Sensitivity* and Table VII.

2) ART-MAPPO Reward-Weight Sensitivity. The sensitivity of the reward design is evaluated in Case 1 using a one-at-a-time perturbation of the five weights in (62). Each weight is scaled by $s \in 0.50, 0.75, 1.00, 1.25, 1.50$, while the remaining weights are fixed at their nominal values. Table VII reports the resulting ISR, temporal-coordination error, terminal lateral acceleration, interception time, and miss distance. The results reveal clear and physically interpretable metric–weight relationships. Increasing w_1 improves terminal accuracy but requires moderately higher terminal lateral acceleration, whereas w_2 has a comparatively mild effect and mainly refines the interception geometry. The terminal-hit weight w_3 primarily governs interception reliability; reducing it to half of its nominal value decreases ISR to 93.8%, while further increasing it provides only marginal reliability improvement at the cost of higher lateral acceleration. The coordination weight w_4 most strongly affects $E_{co\text{-}time}$, which decreases from 0.0830 s to 0.0375 s as the weight increases. In contrast, w_5 directly regulates control effort, reducing E_n from 0.0910 g to 0.0560 g, with a moderate loss in terminal accuracy and interception efficiency. Overall, the nominal configuration provides a balanced operating point among interception reliability, terminal accuracy, temporal coordination, and control economy.

Table VII. Sensitivity Analysis of ART-MAPPO Reward-Function Weighting Coefficients.

Weight	s	ISR (%) ↑	$E_{co\text{-}time}$ (s) ↓	E_n (g) ↓	E_t (s) ↓	E_{miss} (m) ↓
w_1 (dist.)	0.50	96.2	0.0568	0.0675	30.120	3.420
	0.75	97.8	0.0530	0.0696	29.350	3.170
	1.00	**98.7**	**0.0505**	**0.0718**	**28.843**	**2.997**
	1.25	98.9	0.0493	0.0750	28.340	2.860
	1.50	98.8	0.0490	0.0808	28.100	2.830
w_2 (head.)	0.50	97.1	0.0528	0.0785	29.250	3.340
	0.75	98.0	0.0517	0.0748	29.020	3.130
	1.00	**98.7**	**0.0505**	**0.0718**	**28.843**	**2.997**
	1.25	98.8	0.0500	0.0698	28.750	2.890
	1.50	98.7	0.0502	0.0689	28.760	2.870
w_3 (hit)	0.50	93.8	0.0585	0.0645	30.500	3.580
	0.75	97.0	0.0538	0.0687	29.450	3.230
	1.00	**98.7**	**0.0505**	**0.0718**	**28.843**	**2.997**
	1.25	99.0	0.0518	0.0780	28.400	2.900
	1.50	99.1	0.0540	0.0855	28.100	2.870
w_4 (coord.)	0.50	96.8	0.0830	0.0670	28.300	3.150
	0.75	98.0	0.0630	0.0695	28.550	3.060
	1.00	**98.7**	**0.0505**	**0.0718**	**28.843**	**2.997**
	1.25	98.9	0.0430	0.0750	29.050	3.020
	1.50	98.8	0.0375	0.0810	29.550	3.100
w_5 (energy)	0.50	98.9	0.0495	0.0910	28.350	2.880
	0.75	98.8	0.0500	0.0800	28.550	2.930
	1.00	**98.7**	**0.0505**	**0.0718**	**28.843**	**2.997**
	1.25	98.3	0.0518	0.0630	29.200	3.120
	1.50	97.5	0.0545	0.0560	29.850	3.350

Comment 3.6

Reviewer comment 3.6

Moreover, I suggest the author explain whether dual-clip PPO with adaptive KL regularization introduces additional computational overhead during real-time deployment on processors operating in latency constraints within UAV swarms.

Response. We sincerely thank the reviewer for raising this important practical issue. We agree that the computational costs incurred during offline policy optimization should be clearly distinguished from those incurred during real-time execution. In the adopted training-and-deployment framework, Dual-Clip and adaptive KL regularization are evaluated only when optimizing the training loss in Eq. (54). After training, each interceptor executes only the forward pass of its deployed actor; the Dual-Clip operation, KL-divergence estimation, and adaptation of β_{KL} are not evaluated. Therefore, although these mechanisms introduce additional operations during offline training, they add no incremental computation to the real-time inference path. We have clarified this point explicitly in the revised manuscript.

Clarification of real-time computational overhead.

Location of the revision. Section IV-C (*Robust Optimization via Dual-Clip and Adaptive KL*), immediately after the overall ART-MAPPO loss in Eq. (54). The following clarification has been added:

> Dual-Clip and adaptive KL regularization are confined to offline training and are absent from the inference path of the deployed actor. Consequently, they introduce no incremental computational overhead during real-time execution.

Comment 3.7

Reviewer comment 3.7

Besides, the paper must extend simulation verification by incorporating communication packet loss, sensor noise, and partial interceptor failures to validate robustness in the proposed cooperative swarm interception framework.

Response. We sincerely thank the reviewer for this important suggestion. We agree that robustness against target maneuvers alone is insufficient to establish the practical reliability of the proposed framework. The revised evaluation therefore incorporates both information-channel nonidealities and partial interceptor failures.

First, the three-dimensional experimental environment now constructs the defender observations from delayed Cartesian states corrupted by independent Gaussian position and velocity noise. All LOS quantities, closing speed, and time-to-go variables are recomputed from these nonideal measurements. The communication process additionally includes a 1% state-packet-loss rate, with the latest valid state packet retained following a packet loss. These communication and sensing conditions are applied in the reported end-to-end experiments, whose results are presented in Section V (*Simulation Verification*).

Second, we added a dedicated partial-interceptor-failure study. For each of the two engagement cases, 1,000 end-to-end Monte Carlo trials are conducted, with two interceptors randomly designated as failed in every trial. IDBO then constructs the target assignment using the remaining interceptors, after which the surviving swarm executes ART-MAPPO cooperative guidance. Under this reduced interceptor availability, the proposed algorithm maintains interception success rates of 95.4% in Case 1 and 93.9% in Case 2. The corresponding distributions of temporal-coordination error, terminal lateral acceleration, miss distance, and interception time are also reported.

The specific revisions are reproduced below.

(1) Communication packet loss and sensing uncertainty.

Location of the revision. Section V-A (*Experimental Setup and Implementation Details*), in the newly added paragraph *Communication Delay and Sensing Uncertainty*, including Eq. (71).

> *1) Communication Delay and Sensing Uncertainty:* Communication and sensing nonidealities are incorporated into the three-dimensional simulation. The position and velocity measurements used to construct the defender observations are delayed and corrupted by independent zero-mean Gaussian noise,
>
> $$\widehat{\mathbf{p}}(t) = \mathbf{p}(t-\tau_s) + \boldsymbol{\eta}_p, \qquad \widehat{\mathbf{v}}(t) = \mathbf{v}(t-\tau_s) + \boldsymbol{\eta}_v. \tag{71}$$
>
> where $\boldsymbol{\eta}_p \sim \mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I}_3)$ and $\boldsymbol{\eta}_v \sim \mathcal{N}(\mathbf{0}, \sigma_v^2 \mathbf{I}_3)$. The nominal settings are $\tau_s = 50$ ms, $\sigma_p = 1$ m, and $\sigma_v = 0.3$ m/s. All derived observation variables, including the LOS geometry, closing speed, and time-to-go, are recomputed from the delayed noisy Cartesian states. The communication process is subject to a 50 ms communication delay and a 1% packet-loss rate, with the latest valid state packet retained when a packet is lost.

(2) Robustness to partial interceptor failures.

Location of the revision. Section V-F (*Monte Carlo Statistical Analysis and Robustness*), newly added item *(v) Robustness to partial interceptor failures* and Fig. 38.

> *(v) Robustness to partial interceptor failures.* To evaluate robustness to partial interceptor failures, 1,000 end-to-end Monte Carlo trials are conducted for each case. In each trial, two interceptors are randomly designated as failed, and the IDBO module constructs a new assignment using the remaining interceptors. The resulting allocation preserves complete target coverage, after which the surviving interceptor swarm performs cooperative guidance using ART-MAPPO.
>
> Despite the reduced number of available interceptors, the proposed algorithm achieves ISRs of 95.4% in Case 1 and 93.9% in Case 2. The statistical distributions in Fig. 38 further show that the surviving swarm retains terminal accuracy and effective temporal coordination while requiring relatively low terminal lateral acceleration. These results demonstrate that the proposed algorithm maintains stable cooperative interception performance under partial interceptor failures.

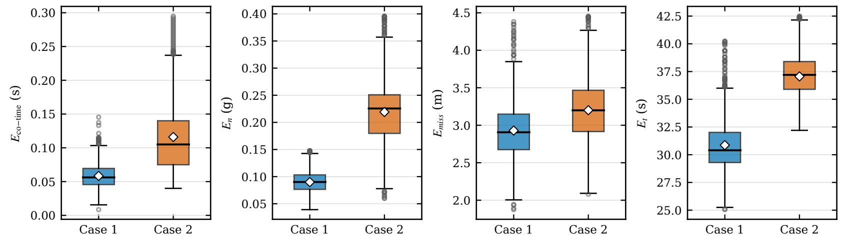

Fig. 38. Performance distributions under two-interceptor failures in Cases 1 and 2.

Comment 3.8

Reviewer comment 3.8

Notably, one of the recommendations is to include failure-case analysis for unsuccessful interception and delayed cooperative engagement scenarios because understanding framework limitations remains essential for practical battlefield deployment reliability.

Response. We sincerely thank the reviewer for this important suggestion. We agree that interception success rates and aggregate statistical metrics alone do not adequately reveal the operating limitations of the proposed framework. We have therefore examined the 13 unsuccessful trials in Case 1 and the 29 unsuccessful trials in Case 2 and added a failure-case analysis to the Monte Carlo discussion.

The analysis distinguishes two failure modes. In an unsuccessful interception, the interceptor swarm begins from an unfavorable engagement geometry in which its initial flight direction differs substantially from the incoming direction of the attacking swarm. Sustained target maneuvering then prolongs trajectory correction and reduces the effective closing rate, so an interceptor may fail to intercept the incoming target before it reaches the high-value target area.

Delayed cooperative engagement represents a different limitation. In this case, the target is eventually intercepted, but differences in interceptor path length and maneuver-induced detours prevent the time-to-go estimates within the corresponding group from converging sufficiently, causing one interceptor to arrive late.

The specific revisions are reproduced below.

Location of the revision. Section V-F (*Monte Carlo Statistical Analysis and Robustness*), within item *(i) Interception reliability and precision*, immediately following the ISR and miss-distance discussion and before item *(ii) Terminal lateral acceleration and operational efficiency*. The revised text is reproduced below:

(i) Interception reliability and precision. The Interception Success Rates (ISRs) directly reflect the algorithms' adaptability to aggressive target maneuvers. While PN achieves a respectable 94.40% against the evasive target (Case 1), its performance precipitously degrades to 17.50% when facing the continuous weaving target (Case 2). Conversely, ART-MAPPO sustains exceptional ISRs of 98.7% and 97.1%, demonstrating profound robustness against continuous, high-frequency line-of-sight (LOS) oscillations. Furthermore, the E_{miss} boxplots confirm that ART-MAPPO consistently achieves a lower median miss distance with a significantly narrower

interquartile range (IQR), ensuring lethal precision even under worst-case initializations. Failure-case analysis was conducted on the 13 unsuccessful trials in Case 1 and 29 in Case 2. For unsuccessful interception, the interceptor swarm starts from an unfavorable engagement geometry, with its initial flight direction differing substantially from the incoming direction of the attacking swarm. Sustained target maneuvering further prolongs the required trajectory correction and reduces the effective closing rate, preventing an interceptor from intercepting the incoming target before it reaches the high-value target area. For delayed cooperative engagement, differences in path length and maneuver-induced detours maintain the time-to-go dispersion within a target group, causing one interceptor to arrive late although the target is eventually intercepted. The former results from the loss of effective closure under unfavorable initial heading geometry and sustained target maneuvering, whereas the latter results from persistent within-group time-to-go differences caused by path asymmetry.

This addition complements the aggregate Monte Carlo results by identifying the specific engagement conditions under which interception reachability or within-group temporal synchronization can deteriorate.

Comment 3.9

Reviewer comment 3.9

In addition, provide a discussion on the generalization capability of the trained ART-MAPPO model under unseen manoeuvring attack patterns, since the current evaluation mainly considers predefined penetration trajectories during testing.

Response. We sincerely thank the reviewer for this important suggestion. We agree that evaluation only under the maneuver patterns represented during model development cannot adequately demonstrate whether the trained policy has learned a transferable interception strategy.

We therefore introduced an additional Case 3 employing bang-bang evasion, which differs from the hybrid evasive maneuver in Case 1 and the continuous sinusoidal maneuver in Case 2. The trained ART-MAPPO policy is kept frozen and applied directly to Case 3 without retraining or fine-tuning; the bang-bang maneuver is not included in its training data. Accordingly, this experiment evaluates zero-shot generalization to an unseen target-maneuver pattern rather than adaptation through additional learning.

A total of 1,000 independent Monte Carlo trials are conducted with randomized initial engagement states. The frozen policy achieves a cooperative interception success rate of 96.4%, a temporal-coordination error of $E_{co\text{-}time} = 0.051$ s, and a miss distance of $E_{miss} = 1.44$ m. The distributions of temporal-coordination error, terminal lateral acceleration, miss distance, and interception time are reported in Fig. 37. These results indicate that ART-MAPPO retains effective interception and within-group coordination under the tested unseen maneuver rather than relying exclusively on the predefined trajectories used during training.

Location of the revision. Section V-F (*Monte Carlo Statistical Analysis and Robustness*), newly added item *(iv) Generalization to unseen attack patterns* and Fig. 37. The corresponding addition is reproduced below:

(iv) Generalization to unseen attack patterns. To further evaluate the proposed method under unseen attack patterns, additional Monte Carlo experiments were conducted in Case 3, where the attackers adopted bang-bang evasion. Over 1,000 independent trials, the proposed method achieves a cooperative interception success rate of 96.4%, with $E_{co\text{-}time} = 0.051$ s and $E_{miss} = 1.44$ m, demonstrating effective temporal coordination and terminal accuracy under this additional maneuvering threat (Fig. 37).

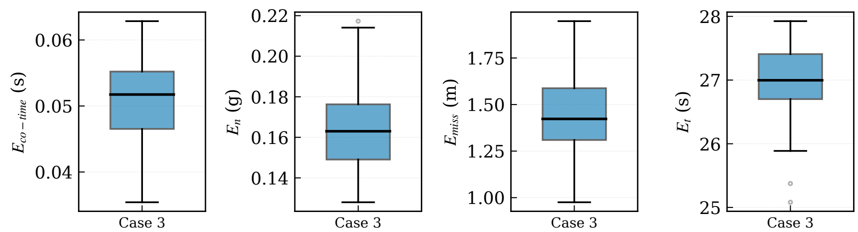

Fig. 37. Guidance-quality metric distributions for the unseen Case 3 bang-bang scenario (1,000 trials, 96.4% ISR): (a) $E_{co\text{-}time}$, (b) E_n, (c) E_{miss}, (d) E_t. Diamonds mark means.

Comment 3.10

Reviewer comment 3.10

At last, it is advisable to incorporate hardware-in-the-loop and semi-physical simulation validation to improve the practical credibility of the ART-MAPPO cooperative tapping strategy for realistic autonomous UAV defence applications effectively.

Response. We sincerely thank the reviewer for this valuable suggestion. We agree that validation based only on a planar mathematical simulation is insufficient to demonstrate the practical implementation capability of the proposed cooperative interception framework.

In response, we first extended the engagement environment from planar motion to three-dimensional dynamics. The revised model includes three-dimensional position propagation, vector-form relative geometry, LOS azimuth and elevation, three-axis acceleration commands, and three-dimensional ZEM.

We then implemented the complete target-assignment and cooperative-guidance workflow on a hardware-in-the-loop semi-physical platform. The policies of defenders D_0–D_4 and their corresponding target-assignment modules are executed on five independent NVIDIA Jetson NX computers. The engagement dynamics, attacking swarm, and software policies of the remaining defenders are executed on the simulation server.

All experiments reported in Section V were conducted on this platform under the communication-delay and sensing-uncertainty environment specified in Section V-A (described in our response to Comment 3.7). The specific revisions are reproduced below.

(1) Three-dimensional engagement model.

Location of the revision. Section II-A (*Problem Modeling*), Eqs. (1)–(3) and (5). The original planar kinematic, relative-motion, acceleration, and ZEM formulations were replaced by their three-dimensional counterparts.

The object of study in this paper is the fixed-wing UAV, and the interception problem is reduced to a two-dimensional horizontal plane formulated in three-dimensional space, as shown in Fig. 2.

$$\begin{cases} \dot{x} = V\cos\theta\cos\gamma, \\ \dot{y} = V\cos\theta\sin\gamma, \\ \dot{z} = V\sin\theta, \end{cases} \tag{1}$$

(x, y, z) are the three-dimensional position coordinates, V is the velocity magnitude, γ is the heading angle, and θ is the flight-path angle.
Let $\mathbf{p}_A, \mathbf{p}_D \in \mathbb{R}^3$ and $\mathbf{v}_A, \mathbf{v}_D \in \mathbb{R}^3$ denote the attacker and defender states. Their three-dimensional relative geometry is

$$\begin{aligned} \mathbf{r}_{AD} &= \mathbf{p}_A - \mathbf{p}_D, \qquad & r &= \|\mathbf{r}_{AD}\|_2, \\ \boldsymbol{\ell}_{AD} &= \mathbf{r}_{AD}/r, \qquad & \dot{r} &= \boldsymbol{\ell}_{AD}^\top(\mathbf{v}_A - \mathbf{v}_D). \end{aligned} \tag{2}$$

where r and $\boldsymbol{\ell}_{AD}$ denote the three-dimensional relative distance and LOS unit vector, respectively. As illustrated in Fig. 2, q_{ij}^{γ} and q_{ij}^{θ} are the corresponding LOS azimuth and elevation angles.

$$\begin{cases} \dot{V} = n_x g, \\ \dot{\gamma} = \dfrac{n_y g}{V\cos\theta}, \\ \dot{\theta} = \dfrac{n_z g}{V}, \end{cases} \tag{3}$$

where n_x, n_y, and n_z denote the axial, lateral, and normal accelerations, and g denotes gravitational acceleration.

$$\mathrm{ZEM} = \left\|\mathbf{r}_{AD} + (\mathbf{v}_A - \mathbf{v}_D)t_{go}\right\|_2. \tag{5}$$

(2) HIL Platform.

Location of the revision. Section V-A (*Experimental Setup and Implementation Details*), in the newly added paragraph *HIL Platform*, together with the platform architecture shown in Fig. 7.

2) HIL Platform: As shown in Fig. 7, the three-dimensional interception experiment is further implemented on a HIL platform. The policies of defenders D_0–D_4 are deployed on five independent NVIDIA Jetson NX computers, with one defender assigned to each node. The corresponding target-assignment algorithm is also executed on the Jetson NX nodes. The engagement dynamics, attacker cluster, and software policies of the remaining defenders D_5–D_n are executed on the simulation server. The server and Jetson NX nodes exchange observations, actions, and target-assignment results through a physical TCP Ethernet network under the communication conditions described above.

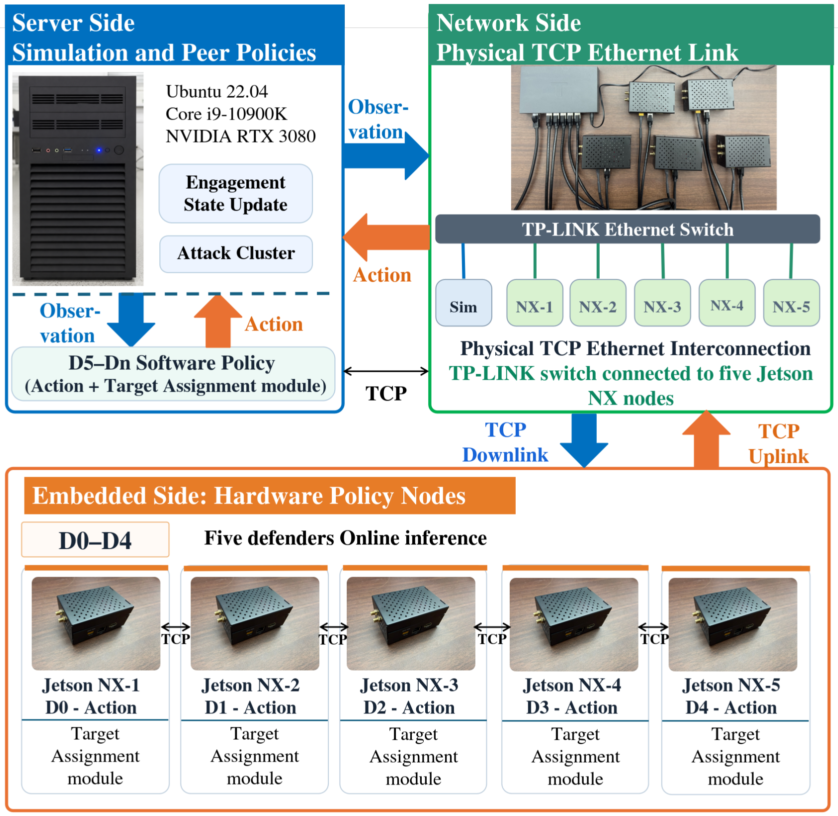

Fig. 7. HIL platform.

These revisions provide a semi-physical closed-loop evaluation in which the proposed target-assignment and ART-MAPPO guidance algorithms run on physical embedded processors and communicate through a physical network, while interacting with a real-time three-dimensional engagement simulation.